

From Data to Models: Analyzing and Integrating Biological Data into Mechanistic Models

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Course of *Doctoral school*
PhD in Complex Systems for Quantitative Biomedicine

When?

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
14	15	16	17	18	19 Sala bianco	20
21	22 Sala bianco	23	24	25	26 Sala bianco	27
28 • 09:00 - 13:00 Sala conferenze, terzo piano 	29	30	1	2	3	4

Course Outline

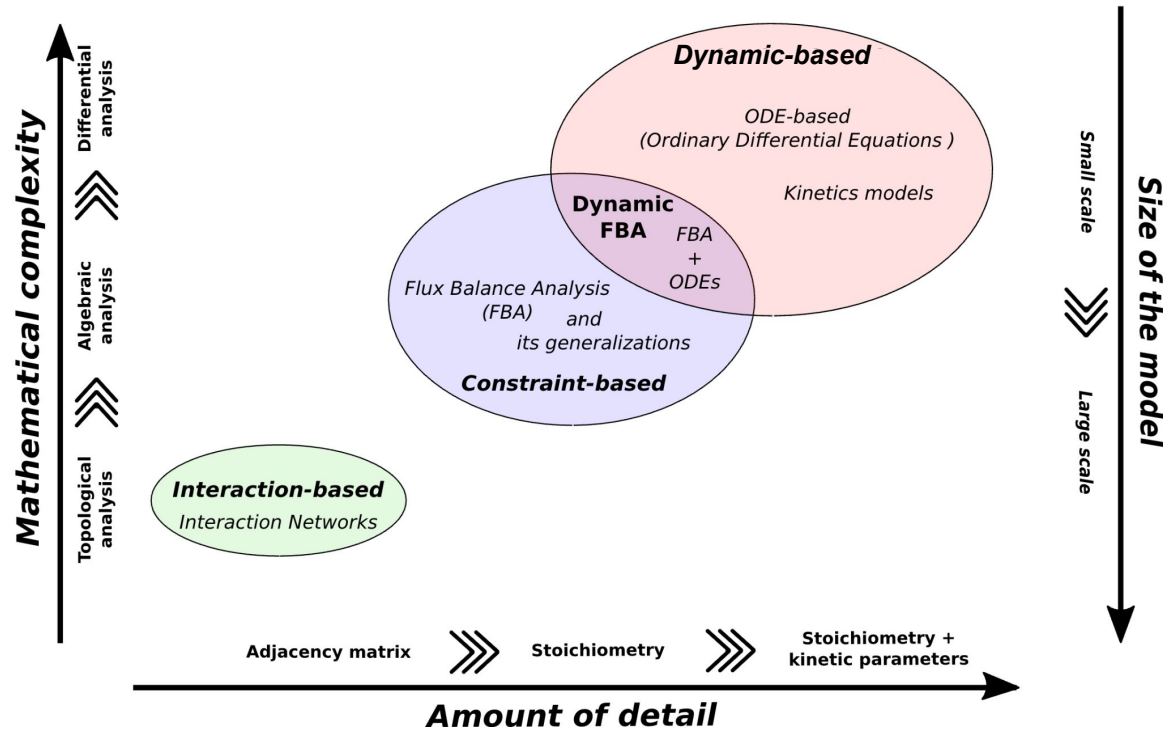
Day 4 – Data Integration and Advanced Modeling

- Flux Balance Analysis: Principles and applications in systems biology.
- Integrative Modeling: combining FBA with experimental data for enhanced predictions.
- Introduction to UnifiedGreatMOD: linking heterogeneous data to mechanistic and constraint-based models.
- Hands-on: FBA case study with data integration.
- Course Wrap-up: Summary of key concepts.



<https://github.com/qBioTurin/From-Data-to-Models.git>

Flux Balance Analysis



Commonly exploited to study systems under specific constraints. Flux Balance Analysis computes reaction fluxes constrained in a range of values, which minimize/maximize an objective function in a system at the equilibrium. Capturing fluxes information in the steady state.

Constraint based models

What Are Constraint-Based Models?

Mathematical models that describe the possible states of a biological system by **imposing constraints** derived from:

- Stoichiometry of biochemical reactions
- Physicochemical laws (e.g., mass conservation, thermodynamics)
- Biological assumptions (e.g., nutrient availability, enzyme capacity)

Instead of trying to model exact dynamics (like ODEs with kinetic parameters), CBMs define a **space of feasible solutions** — all compatible with the known constraints.

Why Use Constraint-Based Modeling?

- **Scalability:** Works at **genome scale** without needing kinetic data.
- **Flexibility:** Can incorporate various biological and environmental conditions.
- **Predictive Power:** Identifies likely cellular behaviors (e.g., growth, secretion) under defined constraints.

Flux Balance Analysis

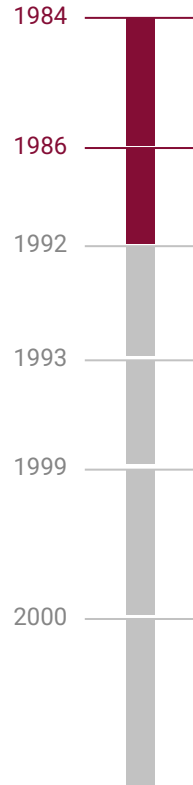
Flux Balance Analysis (FBA) is the most widely used CBM approach, particularly for **steady-state metabolic networks**.

In the early 1990s, the rise of genome sequencing projects and metabolic databases enabled **genome-scale metabolic network reconstructions**.

However, kinetic parameters were largely **unavailable** at this scale

FBA emerged as a **scalable, data-light alternative** to dynamic models.

- Experimental quantification of metabolic fluxes at genome scale is *expensive, time-consuming, and often infeasible*.
- Predict metabolic phenotypes from genomic information.
- Use linear programming (LP) to infer intracellular metabolic states *without requiring detailed kinetic data*.



Papoutsakis(1984) & Papoutsakis and Meyer (1985):

Used LP to calculate maximal theoretical yields

Fell and Small:

Used LP to study lipogenesis

Savinell and Palsson:

Comprehensive assessment of FBA

Varma and Palsson (1993-1995):

Studied metabolic characteristics and capabilities of *E. coli*

Edwards and Palsson (1999-2000):

Gene deletion studies of *E. coli*.
Analysis of *H. Influenzae* metabolic system.
Phenotypic Phase Plane Analysis

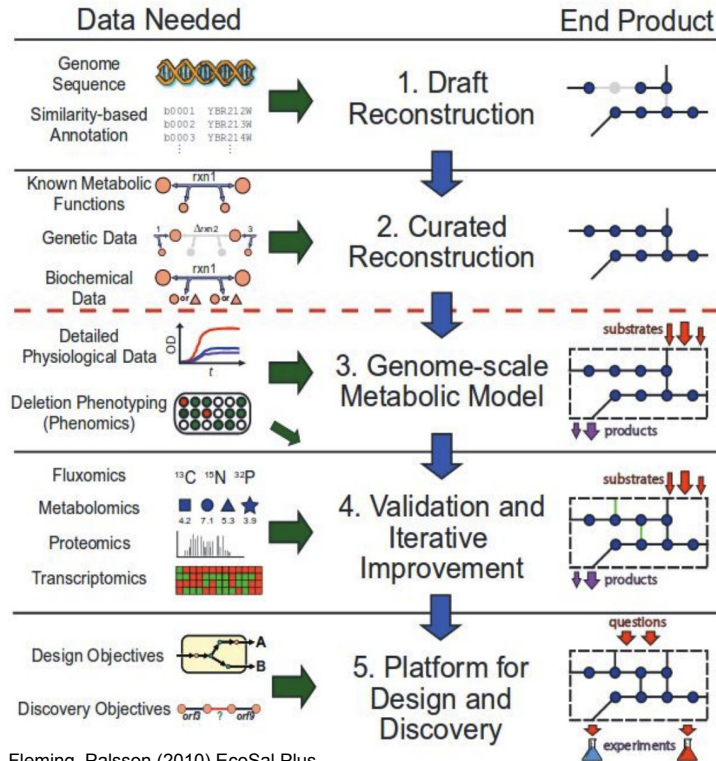
Schilling, Edwards and Palsson:

Integrating Flux-Balance Analysis with Metabolic Pathway Analysis



Genome-scale metabolic model

A **genome-scale metabolic model (GEM)** is a computational representation of all known metabolic reactions in an organism, systematically reconstructed from its annotated genome, biochemical databases, and literature.



How GEMs Are Built

1. **Start assembling the organism's metabolic potential from its DNA:** an initial metabolic network draft, where genes are linked to reactions based on known enzyme functions.
2. **Improve biological accuracy:** a refined model where pathways, reversibility, compartments, and gene-reaction rules (GPRs) are corrected and manually curated.
3. Predict how cells consume substrates and produce products, based on network constraints.
4. Identify and correct model mismatches and increase predictive power.
5. Use the model for metabolic optimization, synthetic biology, drug target discovery, etc

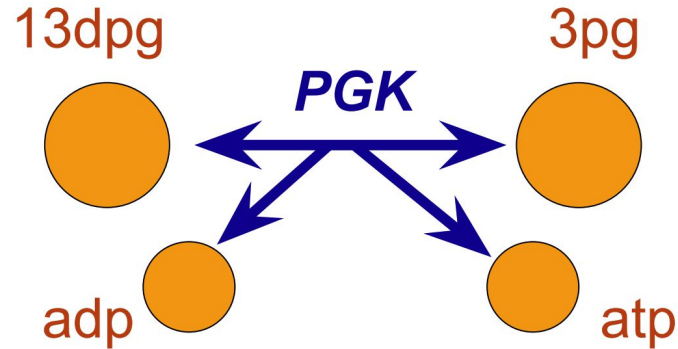
Genome-scale metabolic model

Reactions are associated with specific genes.

In GEMs, each metabolic reaction is linked to one or more specific genes that encode the enzymes catalyzing it.

- **Genes** → produce **proteins** (enzymes)
- **Proteins** → catalyze specific **metabolic reactions**

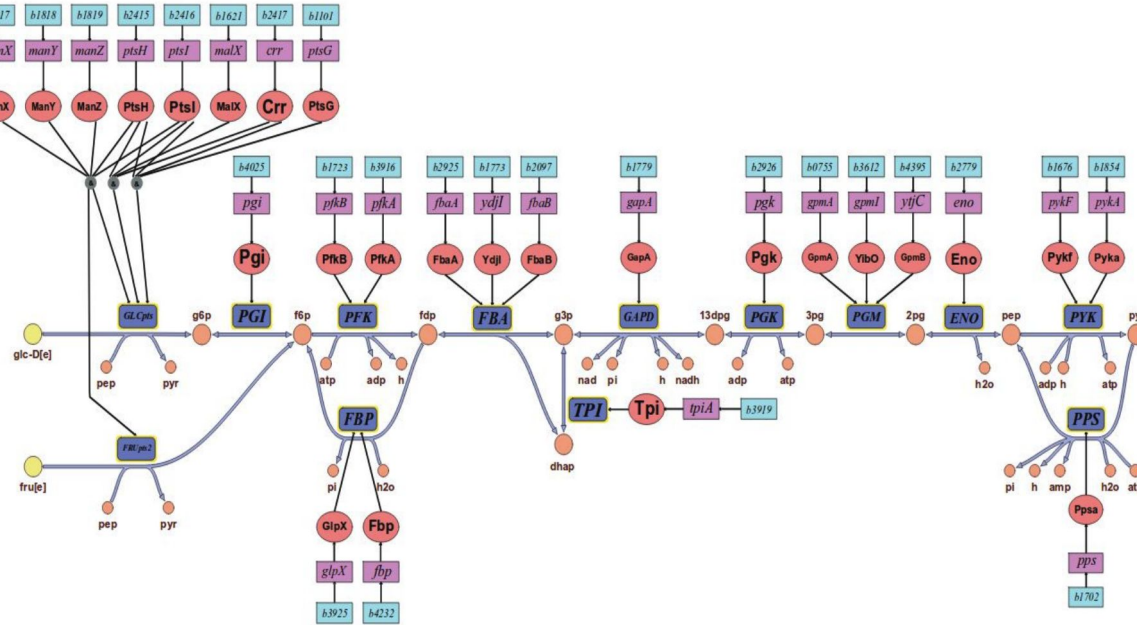
Gene: *pgk*
Protein: Pgk (Phosphoglycerate kinase)
Description: In glycolysis, catalyzes the transfer of a phosphoryl group from 1,3-bisphospho-D-glycerate to ADP, forming ATP and 3-phospho-D-glycerate
Reaction: $13dp\text{g}[c] + \text{adp}[c] \rightleftharpoons 3pg[c] + \text{atp}[c]$



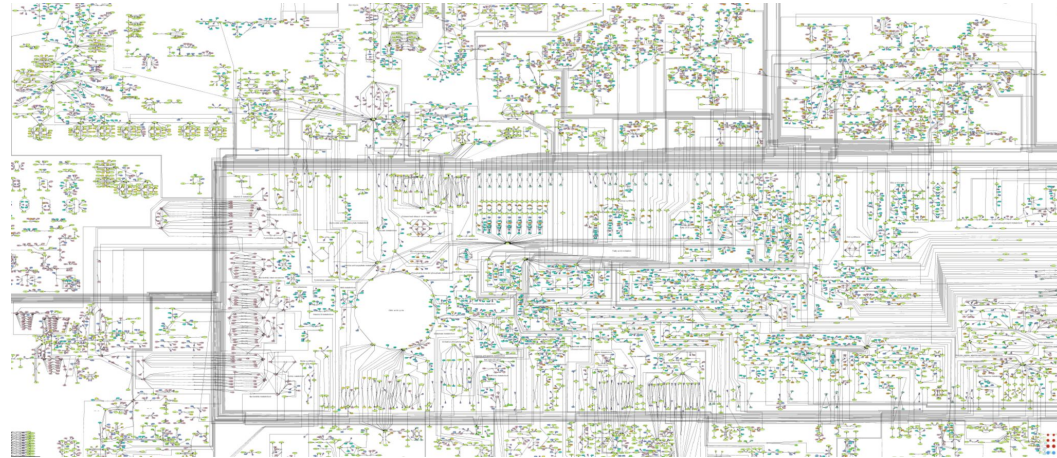
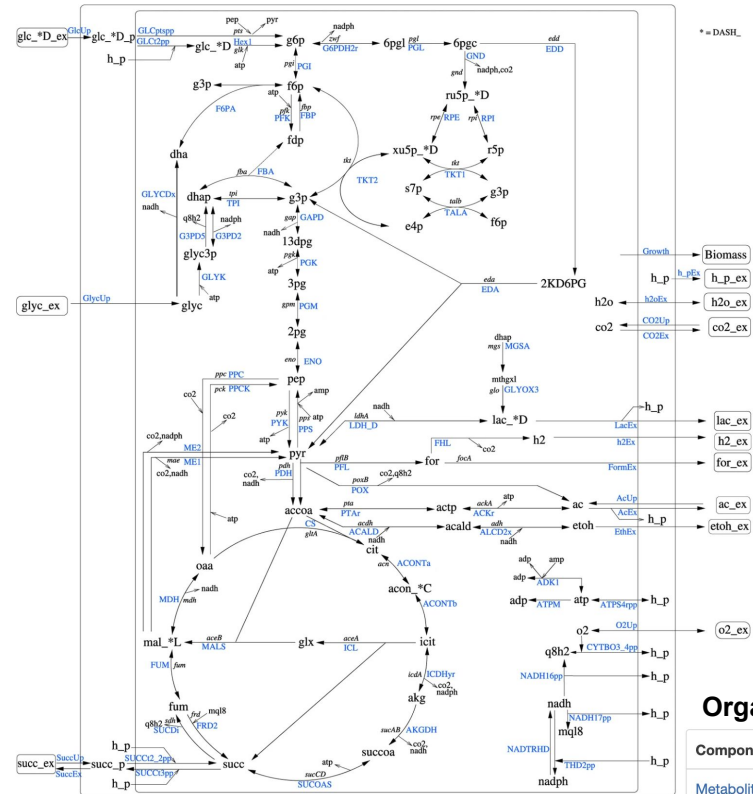
Genome-scale metabolic model

Sets of reactions can be assembled into subsystems

A **subsystem** (also called a **pathway**) is a functional grouping of reactions that together perform a specific biological task: like energy generation, amino acid synthesis, or nucleotide metabolism.



Genome-scale metabolic models



Organism: Homo sapiens

Component	Count
Metabolites	5835
Reactions	10600
Genes	2248

Brunk E, Sahoo S et. al., Nat Biotechnol. (2018)

Organism: Escherichia coli str. K-12 substr. MG1655

Component	Count
Metabolites	1805
Reactions	2583
Genes	1367

Compartments

Compartments are **physiological spaces** inside or around the cell where:

- Reactions take place
- Metabolites are localized
- Transport occurs between regions

Each metabolite in a model is assigned to a **specific compartment** to reflect its **location**.

Symbol	Compartment Name	Description
c	Cytosol	Main intracellular fluid
e	Extracellular space	Outside the cell membrane
m	Mitochondrion	Energy metabolism (TCA cycle, respiration)
r	Endoplasmic reticulum	Protein processing, lipid synthesis
g	Golgi apparatus	Modification and transport
x	Peroxisome	Fatty acid oxidation, detoxification
n	Nucleus	Transcription-related processes
l	Lysosome	Degradation of waste material

glc__D[e] → D-glucose in the **extracellular space**

atp[c] → ATP in the **cytosol**

Online Databases for Metabolic Models

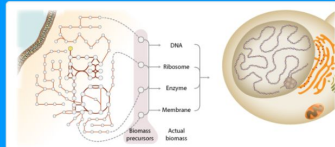
BiGG Models

Search the database by model, reaction, metabolite, or gene ?

Search

☐ Exclude multi-strain models from search

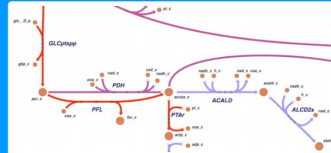
Latest update Version 1.6: Adds 23 new models & more!



View Models



View Metabolites

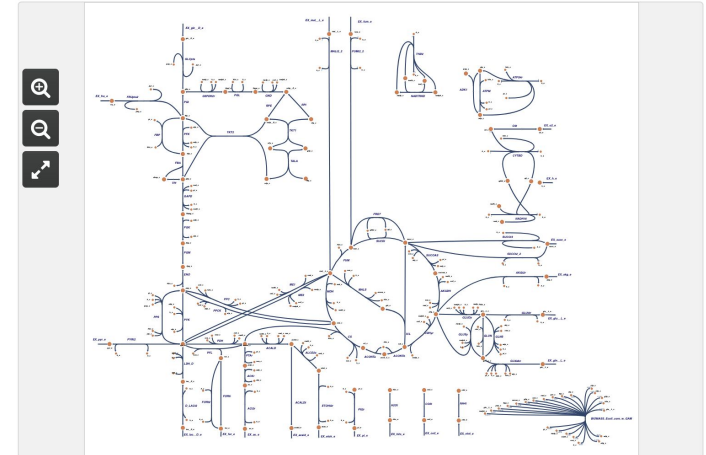


View Reactions

<http://bigg.ucsd.edu/>

Visualize models online with:

Escher Map ?



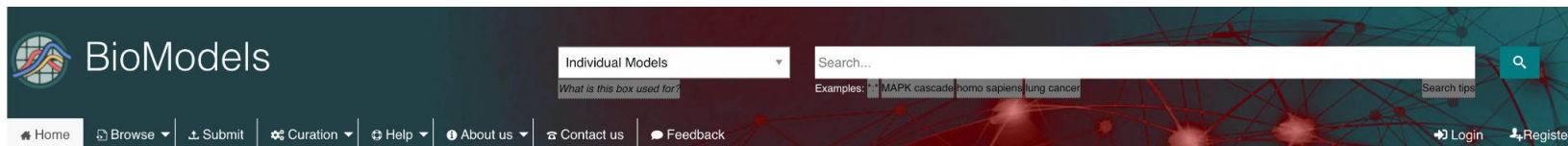
Open map in a new window

Escher homepage

http://bigg.ucsd.edu/models/e_coli_core

Repository of **GEMs** for many organisms (e.g., *E. coli*, human, yeast).

Online Databases for Metabolic Models



BioModels is a repository of mathematical models of biological and biomedical systems. It hosts a vast selection of existing literature-based physiologically and pharmaceutically relevant mechanistic models in standard formats. Our mission is to provide the systems modelling community with reproducible, high-quality, freely-accessible models published in the scientific literature. More information about using BioModels such as [model submission](#), [update](#), [publication](#), or [reviewers access](#) can be found in the [FAQ](#).



Submission / Update



Manually Curated

1,096 models



Non-curated

1,648 models



Auto generated

833 models



BioModels Parameters

899,648 records

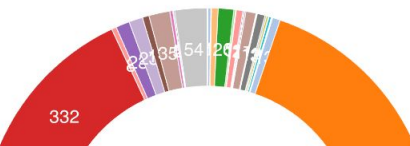
Model of The Month



July, 2025

Basal MET Phosphorylation as an Indicator of Hepatocyte Dysregulation in Liver Disease

In our recent study, conducted in the frame of the LiSyM-Cancer Network funded by the German Federal Ministry for Education and Research, we sought to unravel how high-fat, high-sugar diets influence intracellular signalling in hepatocytes. Using a dynamic mathematical model of hepatocyte growth factor (HGF) signal transduction, we identified the basal phosphorylation of the MET receptor as a key indicator of hepatocyte dysregulation.



Browse by Modelling Approach

This shows models distribution based on modelling approaches. Click on a slice to display models.

Find us on Twitter

[Tweets by @biomodels](#)

Repository of thousands of computational models (dynamics + constraint-based).

<https://www.ebi.ac.uk/biomodels/>

Online Databases for Metabolic Models

metabolic
ATLAS





Explore

GEM 

GotEnzymes

Documentation

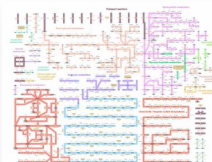
About 

METABOLIC ATLAS

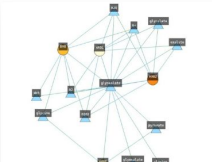
open source genome-scale metabolic models for easy browsing and analysis



GEM Browser



Map Viewer



Interaction Partners

explore the integrated models

1. Select a model:

☐ **Fruitfly-GEM 1.3.0**
Drosophila melanogaster

11873 reactions - 8092 metabolites - 1753 genes

☒ **Human-GEM 1.14.0**
Homo sapiens

13024 reactions - 8363 metabolites - 2920 genes

☐ **Mouse-GEM 1.4.0**
Mus musculus

13018 reactions - 8367 metabolites - 2859 genes

☐ **Rat-GEM 1.4.0**
Rattus norvegicus

13026 reactions - 8371 metabolites - 2818 genes

☐ **Worm-GEM 1.4.0**
Caenorhabditis elegans

12020 reactions - 8127 metabolites - 1615 genes

☐ **Yeast-GEM 8.6.2**
Saccharomyces cerevisiae

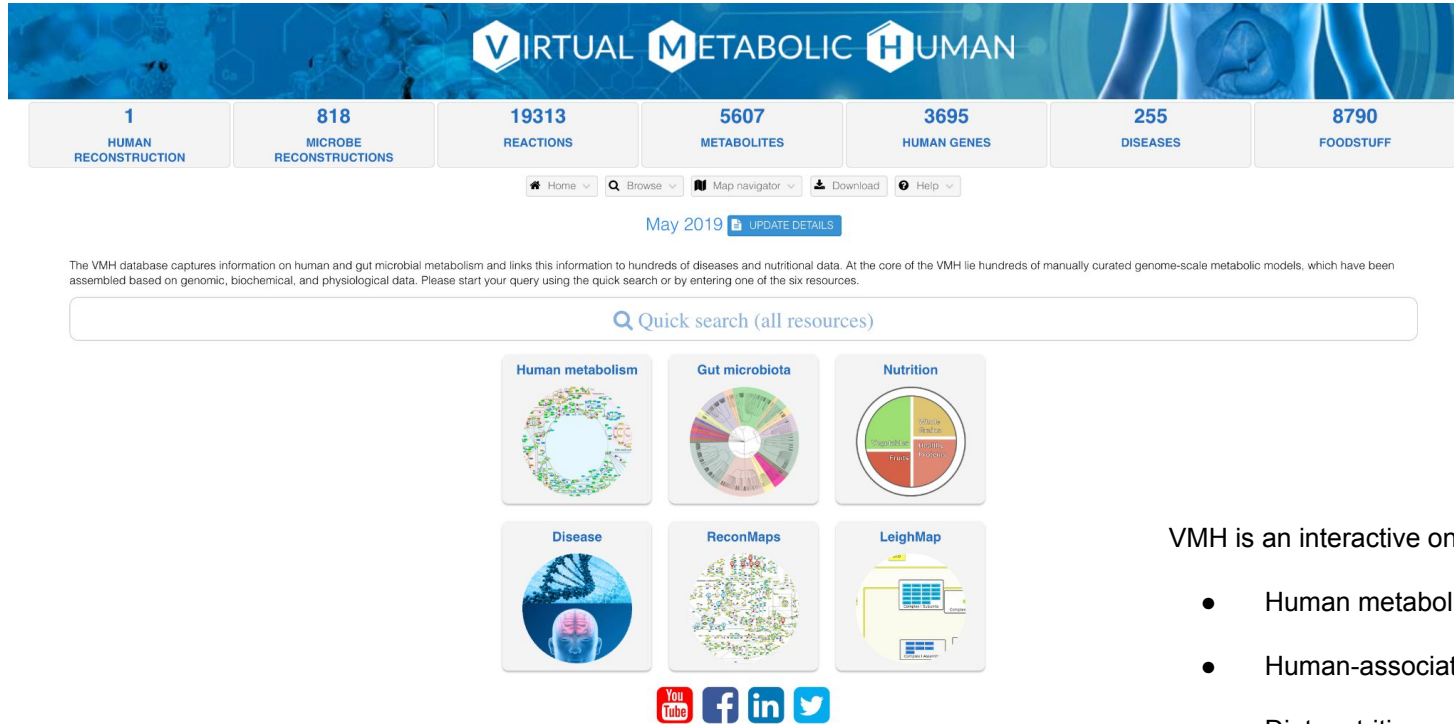
4063 reactions - 2744 metabolites - 1160 genes

☐ **Zebrafish-GEM 1.3.0**
Danio rerio

12868 reactions - 8353 metabolites - 2736 genes

<https://metabolicatlas.org/>

Online Databases for Metabolic Models



<https://www.vmh.life/#home>

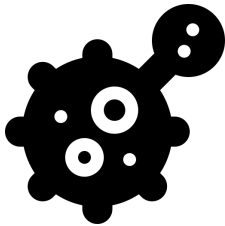
VMH is an interactive online platform for exploring:

- Human metabolism
- Human-associated microbes (including *E. coli*)
- Diet, nutrition, and host–microbe interactions
- Genome-scale metabolic reconstructions (GEMs)

Why study metabolism?

Metabolism is the network of chemical reactions that sustains life.

- Produces **energy** (ATP, NADH)
- Synthesizes **biomolecules** (amino acids, nucleotides, lipids)
- Maintains **redox and pH balance**
- Supports **growth, reproduction, and adaptation**



Applications in Bioengineering:

Metabolic engineering optimizes organisms to produce **fuels, drugs, and food**.

Significance in Medicine:

Many diseases are fundamentally **metabolic**

Tumor metabolism: Warburg effect

From Metabolism to Modeling

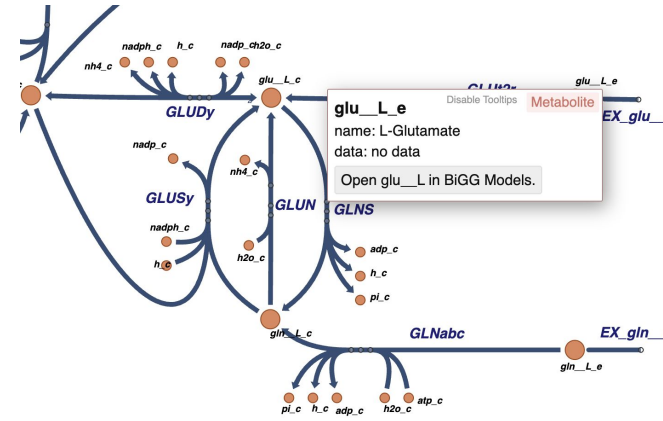
What Are Metabolites?

Small molecules involved in or produced by metabolic reactions

Metabolites are the variables of the system.

Metabolite Abbreviation	Metabolite Name	Neutral Formula	Charged Formula	Charge	Compartment	KEGG ID	CAS Number	Alternate Names
10fth[c]	10-Formyltetrahydrofolate	C20H23N7O7	C20H21N7O7	-2	Cytosol	C00234		10-Formyl-THF
12dgr120[c]	1,2-Diacyl-sn-glycerol (didodecanoyl, n-C12:0)	C27H52O5	C27H52O5	0	Cytosol	C00641		1,2-Diacylglycerol/ D-1,2-Diacylglycerol
12dgr140[c]	1,2-Diacyl-sn-glycerol (ditetradecanoyl, n-C14:0)	C31H60O5	C31H60O5	0	Cytosol	C00641		D-1,2-Diacylglycerol/ 1,2-Diacylglycerol
12dgr141[c]	1,2-Diacyl-sn-glycerol (ditetradec-7-enoyl, n-C14:1)	C31H56O5	C31H56O5	0	Cytosol	C00641		D-1,2-Diacylglycerol/ 1,2-Diacylglycerol
12dgr160[c]	1,2-Diacyl-sn-glycerol (dihexadecanoyl, n-C16:0)	C35H68O5	C35H68O5	0	Cytosol	C00641		D-1,2-Diacylglycerol/ 1,2-Diacylglycerol
12dgr161[c]	1,2-Diacyl-sn-glycerol (dihexadec-9-enoyl, n-C16:1)	C35H64O5	C35H64O5	0	Cytosol	C00641		D-1,2-Diacylglycerol/ 1,2-Diacylglycerol
12dgr180[c]	1,2-Diacyl-sn-glycerol (dioctadecanoyl, n-C18:0)	C39H76O5	C39H76O5	0	Cytosol	C00641		D-1,2-Diacylglycerol/ 1,2-Diacylglycerol
12dgr181[c]	1,2-Diacyl-sn-glycerol (dioctadec-11-enoyl, n-C18:1)	C39H72O5	C39H72O5	0	Cytosol	C00641		1,2-Diacylglycerol/ D-1,2-Diacylglycerol
12ppd-R[c]	(R)-Propane-1,2-diol	C3H8O2	C3H8O2	0	Cytosol	C02912	4254-14-2	(R)-1,2-Propanediol/ (R)-Propylene glycol/ D
12ppd-S[c]	(S)-Propane-1,2-diol	C3H8O2	C3H8O2	0	Cytosol	C02917	4254-14-3	(S)-1,2-Propanediol/ (S)-Propylene glycol/ L
13dpg[c]	3-Phospho-D-glyceroyl phosphate	C3H8O10P2	C3H4O10P2	-4	Cytosol	C00236		1,3-bis-phosphoglycerate/ 3-Phospho-D-gly
14dhnc[a][c]	1,4-dihydroxy-2-naphthoyl-CoA	C32H38N7O19P3S	C32H38N7O19P3S	-4	Cytosol	C15547		
14glucan[c]	1,4-alpha-D-glucan	C36H62O31	C36H62O31	0	Cytosol	C00718		

Main Attribute	Description	Example
Name	Human-readable name of the metabolite	"Glucose"
Description	Optional longer description	"D-Glucose in cytosol"
Charged Formula	Chemical formula including protonation state	"C6H12O6" or "C6H11O6"
Compartment	Location within the cell or environment	c = cytosol, e = extracellular, m = mitochondria



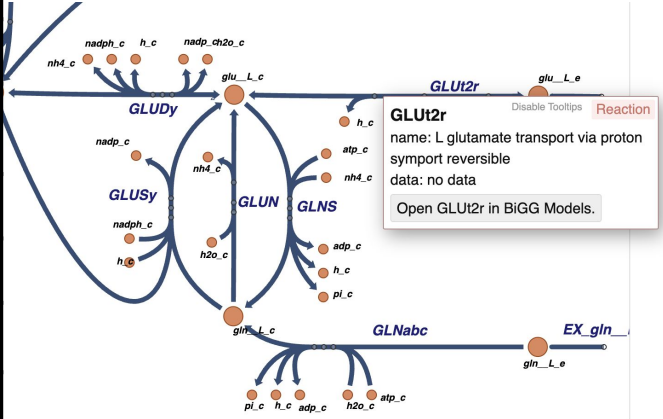
From Metabolism to Modeling

What Are Metabolic Reactions?

A process where two or more reactant molecules interact, typically facilitated by an enzyme, to form a product

Name	Description	Formula	Gene-Protein-Reaction Association	Gene-Reaction Association	Protein-Reaction Association
ENO	enolase	2pgl[c] <=> h2o[c] + pep[c]	Eno (b2779)	b2779	Eno
F6PA	fructose 6-phosphate aldolase	f6p[c] <=> dhap[c] + g3p[c]	(Fsa (b0825)) or (TalC (b3946))	(b0825 or b3946)	(Fsa) or (TalC)
FBA	fructose-bisphosphate aldolase	fdp[c] <=> dhap[c] + g3p[c]	(FbaB (b2097)) or (B1773 (b1773)) or (b2097 or b1773 or b2925)	(FbaB) or (B1773) or (b2097 or b1773 or b2925)	(FbaB) or (B1773) or (b2097 or b1773 or b2925)
FBP	fructose-bisphosphatase	fdp[c] + h2o[c] -> f6p[c] + pi[c]	(GlpX (b3925)) or (Fbp (b4232)) or (b3925 or b4232 or b2930)	(GlpX) or (Fbp) or (b3925 or b4232 or b2930)	(GlpX) or (Fbp) or (b3925 or b4232 or b2930)
G1PPpp	Glucose-1-phosphatase	g1p[p] + h2o[p] -> glc-D[p] + pi[p]	Agp (b1002)	b1002	Agp
G6PP	glucose-6-phosphate phosphatase	g6p[c] + h2o[c] -> glc-D[c] + pi[c]	YbiV (b0822)	b0822	YbiV
ICAD	nicotinic acid deaminase	nic[c] + h2o[c] -> pyr[c] + h[c] + nadh[c]	GapA (b1779)	b1779	GapA

Attribute	Description	Example
Name	Human-readable reaction name	"Hexokinase reaction"
ID	Unique identifier in the model	HEX1 or rxn00001
Description	Optional long description	"Phosphorylation of glucose to G6P"
Formula	The stoichiometric equation	glc__D_c + atp_c -> g6p_c + adp_c + h_c
Gene-Protein-Reaction Association	Logical rule linking gene(s) to the enzyme catalyzing the reaction	(b2388 and b3916) or (geneA or geneB)
Protein(s)	The enzyme(s) that catalyze the reaction	"Hexokinase"
Gene(s)	Specific gene(s) encoding the protein(s)	b2388, b3916 (E. coli locus tags)
Cellular Subsystem	Functional grouping of the reaction	"Glycolysis", "TCA cycle"
Compartment(s)	Cellular location of the reaction	c = cytosol, m = mitochondria
Flux Bounds	Allowed range for reaction flux	Lower: 0, Upper: 1000 (if irreversible)



From Metabolism to Modeling

Why are the metabolites attributes important?

- Needed for reaction balancing and mass-charge consistency
- Helps distinguish isomers, localizations (e.g. NADH_c vs. NADH_m)
- **Conversion/Export:** Required when exporting models

Why are the reactions attributes matter?

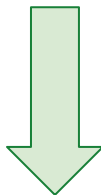
- **Flux bounds** determine the feasible solution space
- **GPR rules** allow for gene knockouts or overexpression simulations
- **Subsystems** help organize large networks functionally
- **Compartment info** is essential for mass balance and transport modeling



From Metabolism to Modeling

Dynamics of metabolic behavior over time:

- Metabolite concentrations;
- Enzyme concentrations;
- Enzyme activity rate depends on enzyme concentrations and metabolite concentrations;
- Solved using a set of differential equations.



Modeling the Network Function: Kinetic Models

Impossible to model large-scale networks:

Requires specific enzyme rates data;
Too complicated to obtain.



Flux Balance Analysis

FBA was conceived to address key challenges in metabolic modeling:

- **Lack of detailed kinetic data:** modeling large, genome-scale networks with ODEs is unfeasible without comprehensive reaction kinetics.
- **Absence of dynamic measurements:** intracellular metabolite concentrations are often unavailable or unreliable.
- **Computational efficiency:** FBA enables rapid predictions of metabolic phenotypes using only stoichiometry and constraints.
- **Optimization hypothesis:** by using a biologically relevant objective (e.g., growth), one can predict cellular behavior under different conditions.



A computational method to **predict metabolic fluxes** through a network **under steady-state conditions**, using **linear optimization**.

Although FBA does **not** predict **metabolite concentrations**, the **fluxes it predicts are biologically meaningful** and useful for many types of analyses and applications.

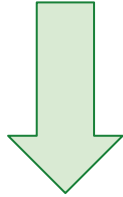
Predicting Cell Growth:

The biomass reaction flux approximates the growth rate which is useful in: Nutrient testing (glucose vs. glycerol), Antibiotic conditions, Environmental adaptation (aerobic vs anaerobic)

Flux Balance Analysis

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A computational method to **predict metabolic fluxes** through a network **under steady-state conditions**, using **linear optimization**.

Core Assumptions:

Steady State: Intracellular metabolite concentrations don't change over time

Mass Balance: What comes into each metabolite = what goes out

Constraints: Nutrient availability (e.g., glucose uptake), Thermodynamics (reaction directionality), Flux capacity (enzyme limitations)

Objective Function: Cell optimizes a biological goal — often **biomass production**

The steady state assumption

Let's consider a system:

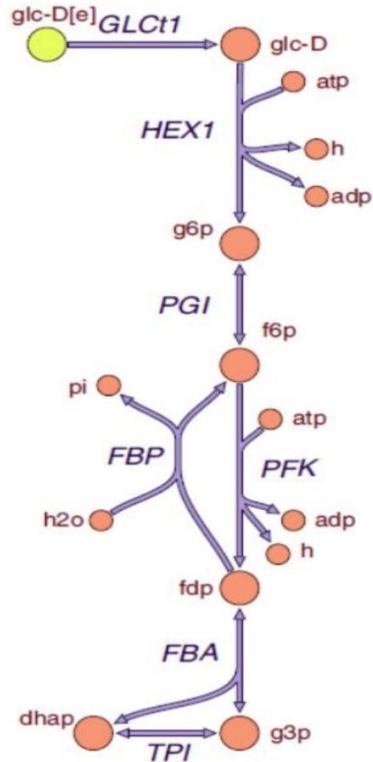
1. r reactions
2. m metabolites
3. $\mathbf{x}$ the vector of the metabolite concentrations
4. $\mathbf{v}$ the vector of the flux velocities of each reaction

The inner product of the **stoichiometric matrix** S and the **flux vector** $\mathbf{v}$ gives the **change in metabolite concentrations over time**, where $\mathbf{x}$ is a vector of metabolite concentrations.

$$\frac{d\mathbf{x}}{dt} = S \cdot \mathbf{v}$$

The steady state assumption

Simplified glycolytic pathway



Stoichiometric Matrix (S-matrix)

	GLCt1	HEX1	PGI	PFK	FBP	FBA	TPI	EX_glc
glc-D[e]	-1	0	0	0	0	0	0	-1
glc-D	1	-1	0	0	0	0	0	0
atp	0	-1	0	-1	0	0	0	0
H	0	1	0	1	0	0	0	0
adp	0	1	0	1	0	0	0	0
g6p	0	1	-1	0	0	0	0	0
f6p	0	0	1	-1	1	0	0	0
fdp	0	0	0	1	-1	-1	0	0
pi	0	0	0	0	1	0	0	0
h2o	0	0	0	0	-1	0	0	0
g3p	0	0	0	0	0	1	1	0
dhap	0	0	0	0	0	1	-1	0

Each cell (i,j) is the **stoichiometric coefficient** of metabolite *i* in reaction *j*.

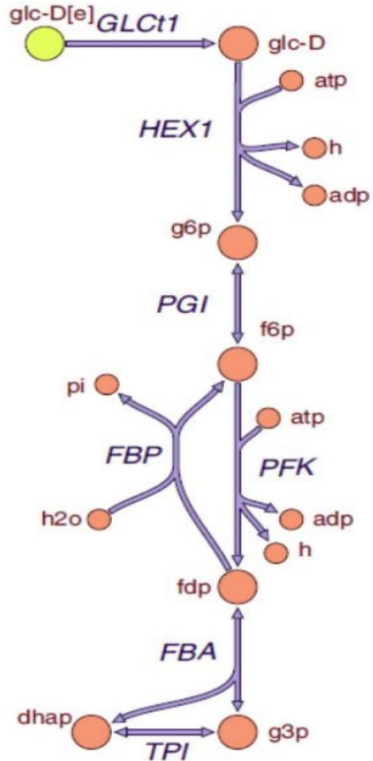
Negative value (-1): Metabolite is consumed (reactant)

Positive value (+1): Metabolite is produced (product)

0: Metabolite is not involved in that reaction

The steady state assumption

Simplified glycolytic pathway



Stoichiometric Matrix (S-matrix)

	GLCt1	HEX1	PGI	PFK	FBP	FBA	TPI	EX_glc
glc-D[e]	-1	0	0	0	0	0	0	-1
glc-D	1	-1	0	0	0	0	0	0
atp	0	-1	0	-1	0	0	0	0
H	0	1	0	1	0	0	0	0
adp	0	1	0	1	0	0	0	0
g6p	0	1	-1	0	0	0	0	0
f6p	0	0	1	-1	1	0	0	0
fdp	0	0	0	1	-1	-1	0	0
pi	0	0	0	0	1	0	0	0
h2o	0	0	0	0	-1	0	0	0
g3p	0	0	0	0	0	1	1	0
dhap	0	0	0	0	0	1	-1	0

exchange reaction:
it brings glucose into the system

Each cell (i,j) is the **stoichiometric coefficient** of metabolite *i* in reaction *j*.

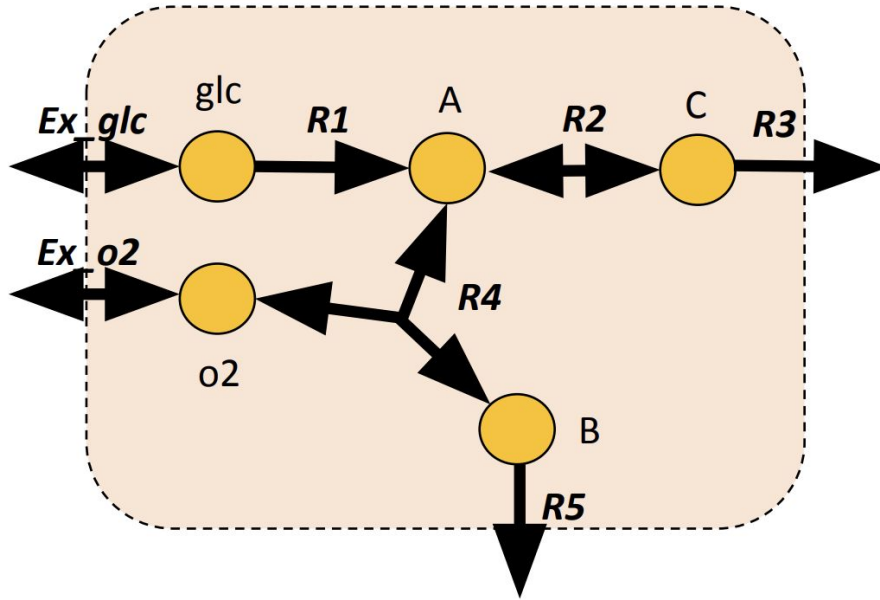
Negative value (-1): Metabolite is consumed (reactant)

Positive value (+1): Metabolite is produced (product)

0: Metabolite is not involved in that reaction

The steady state assumption

Which is the S for the following system?

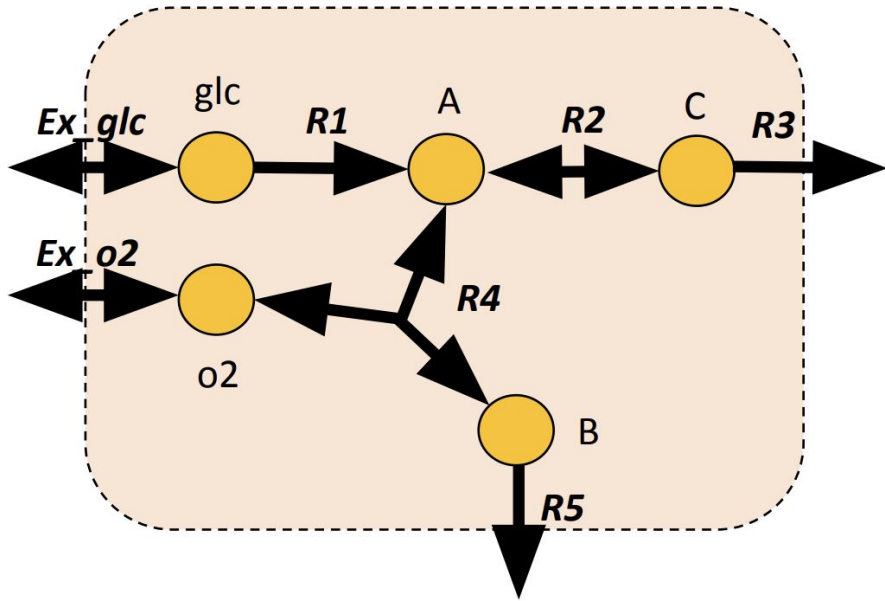


Reactions

<i>Ex_glc</i>	$\text{glc} \rightleftharpoons 0$
<i>Ex_o2</i>	$\text{o2} \rightleftharpoons 0$
<i>R1</i>	$\text{glc} \rightarrow \text{A}$
<i>R2</i>	$\text{A} \rightleftharpoons \text{C}$
<i>R3</i>	$\text{C} \rightarrow 0$
<i>R4</i>	$\text{A} + \text{o2} \rightleftharpoons \text{B}$
<i>R5</i>	$\text{B} \rightarrow 0$

The steady state assumption

Which is the S for the following system?



Metabolite	Ex_glc	Ex_o2	R1	R2	R3	R4	R5
glc	-1	0	-1	0	0	0	0
o2	0	-1	0	0	0	-1	0
A	0	0	+1	-1	0	-1	0
C	0	0	0	+1	-1	0	0
B	0	0	0	0	0	+1	-1

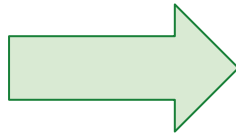
The steady state assumption

Let's consider a system:

1. r reactions
2. m metabolites
3. $\mathbf{x}$ the vector of the metabolite concentrations
4. $\mathbf{v}$ the vector of the flux velocities of each reaction

The inner product of the **stoichiometric matrix** $\mathbf{S}$ and the **flux vector** $\mathbf{v}$ gives the **change in metabolite concentrations over time**, where $\mathbf{x}$ is a vector of metabolite concentrations.

$$\frac{d\mathbf{x}}{dt} = \mathbf{S} \cdot \mathbf{v}$$



The cell is in a **pseudo-equilibrium**, where metabolite pools are stable over the timescale of interest.

Assuming the cell is in one phenotype for a time longer than it takes for metabolite concentrations to change dramatically

$$\frac{d\mathbf{x}}{dt} = 0 = \mathbf{S} \cdot \mathbf{v}$$

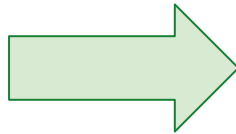
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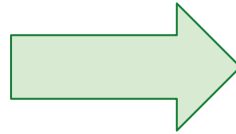
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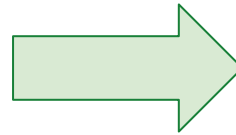
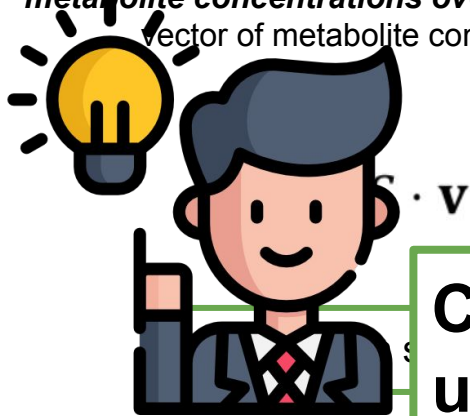
There are many more reactions (unknown variables) than metabolites, there will not be one unique solution

The steady state assumption

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2. m metabolites
3. x the vector of the metabolite concentrations
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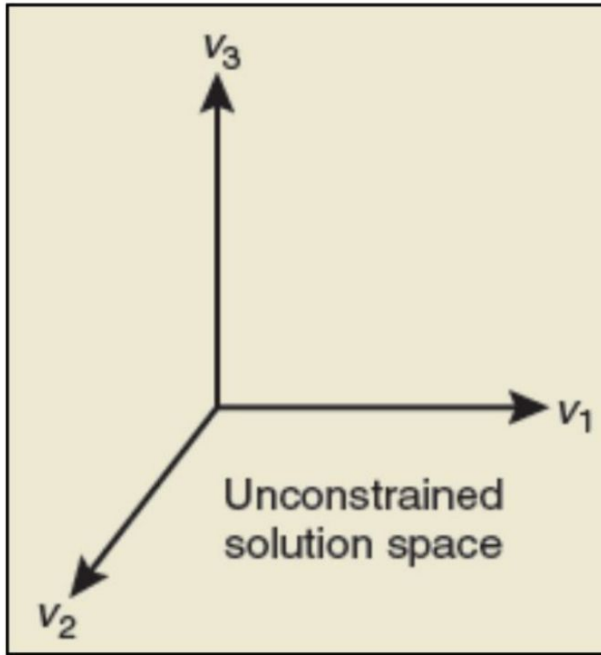
Assuming the cell is in one phenotype for a time longer than it takes for metabolite concentrations to change dramatically

$$\frac{dx}{dt} = 0 = S \cdot v$$

Constraining the solution space and using linear programming

(es) than

Constraining the solution space

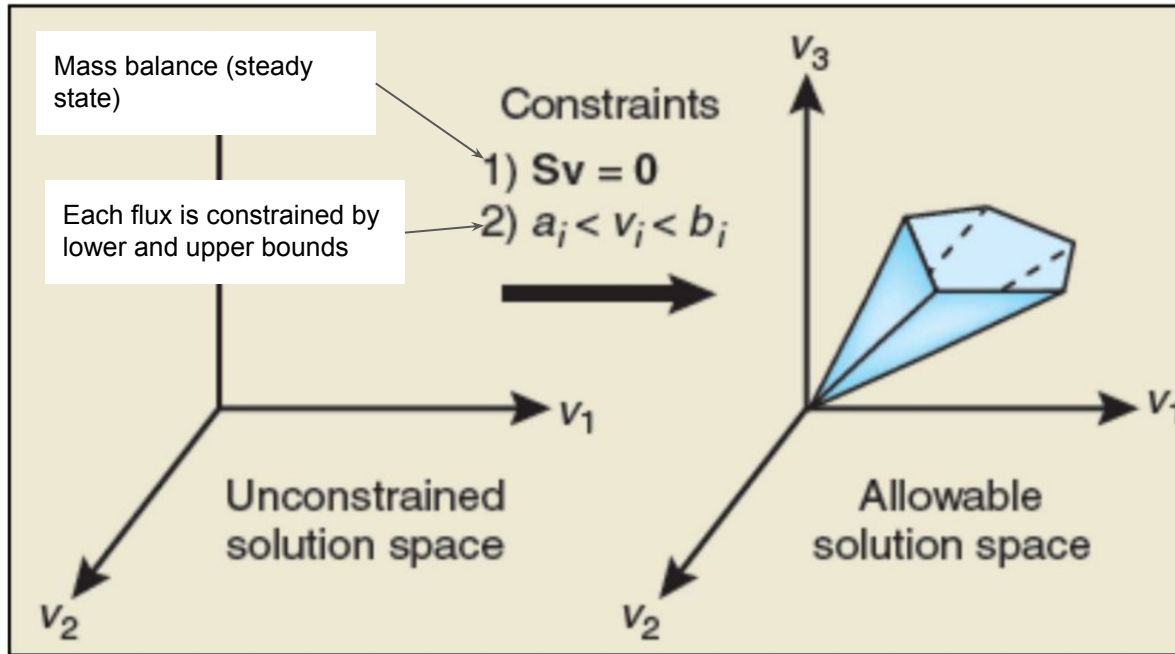


We have **three fluxes**: v_1 , v_2 , and v_3 .

In theory, each can take **any value**, so the space of all possible combinations is infinite.

But this is **biologically unrealistic**.

Constraining the solution space



These constraints define a (feasible) ***allowable solution space***

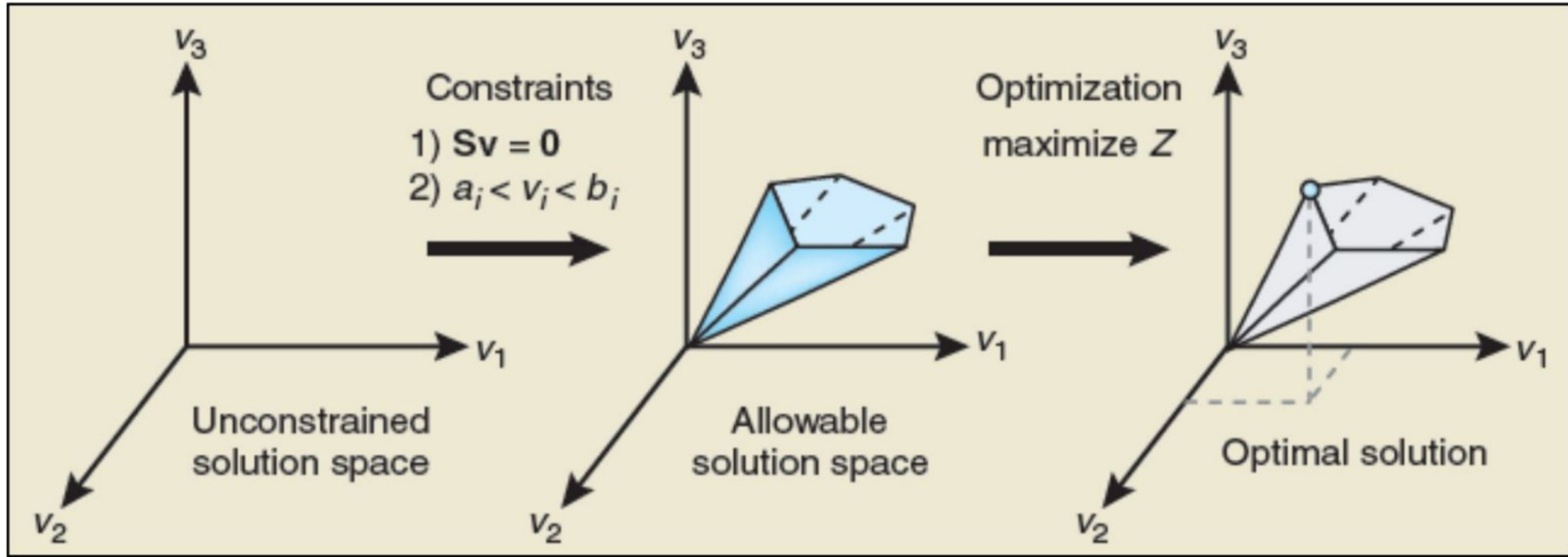
Flux constraints:

Irreversibility (e.g., $v \geq 0$)

Maximum uptake or secretion limits

Biological feasibility

Constraining the solution space



Example 2

Consider a system that has two metabolites A and B.

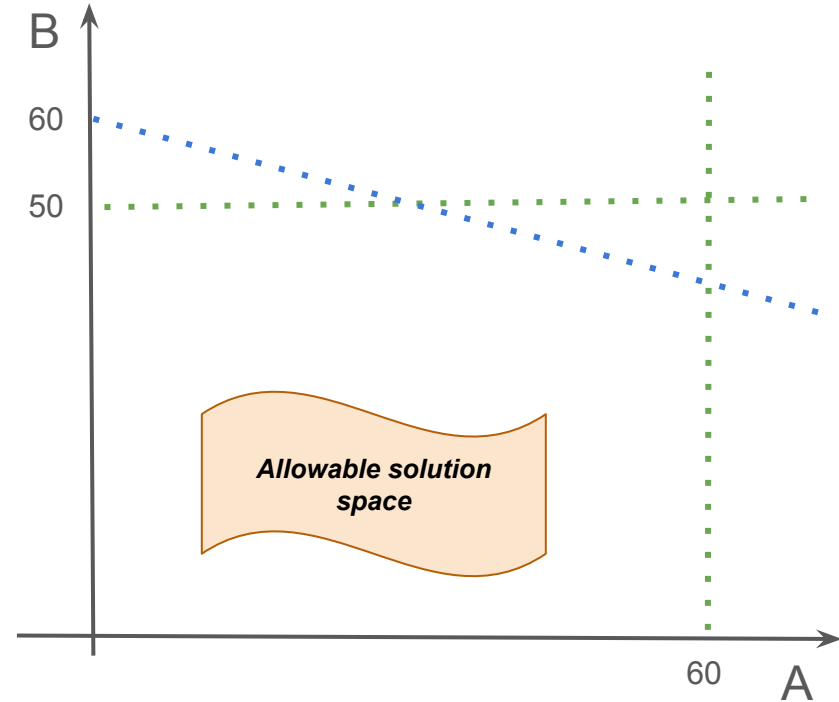
The production constraints on them are:

$$0 < A < 60$$

$$0 < B < 50$$

Additionally, the capacity for producing them simultaneously is limited by:

$$A + 2B < 120$$



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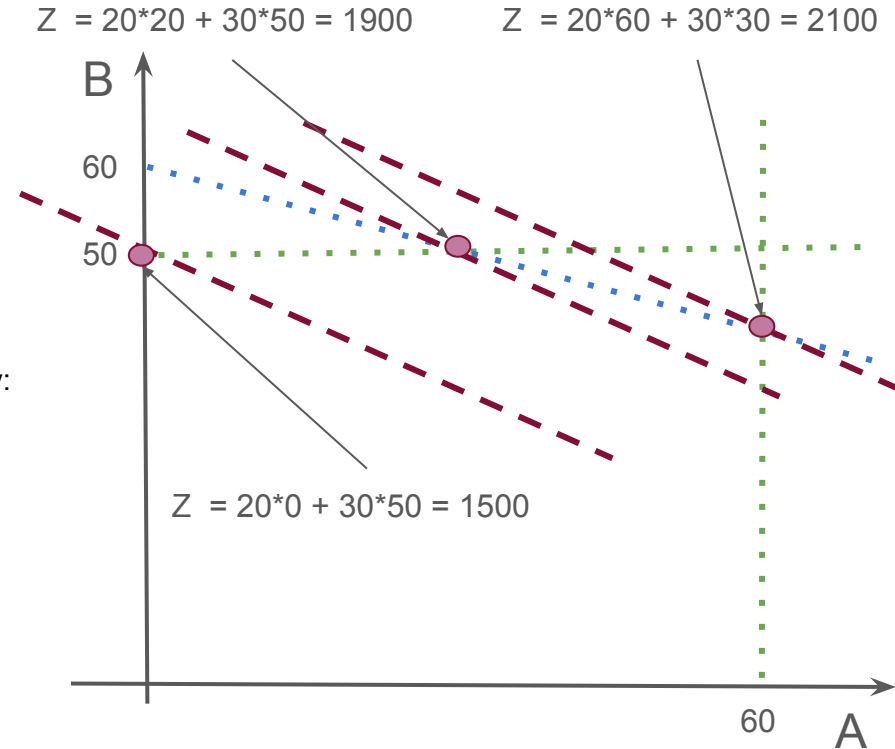
$$0 < B < 50$$

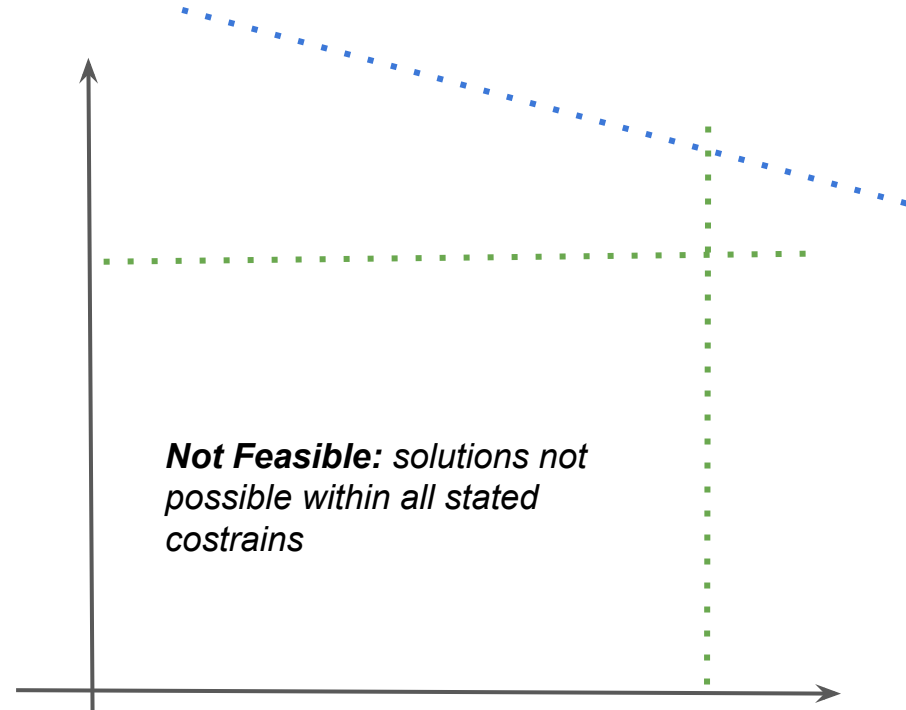
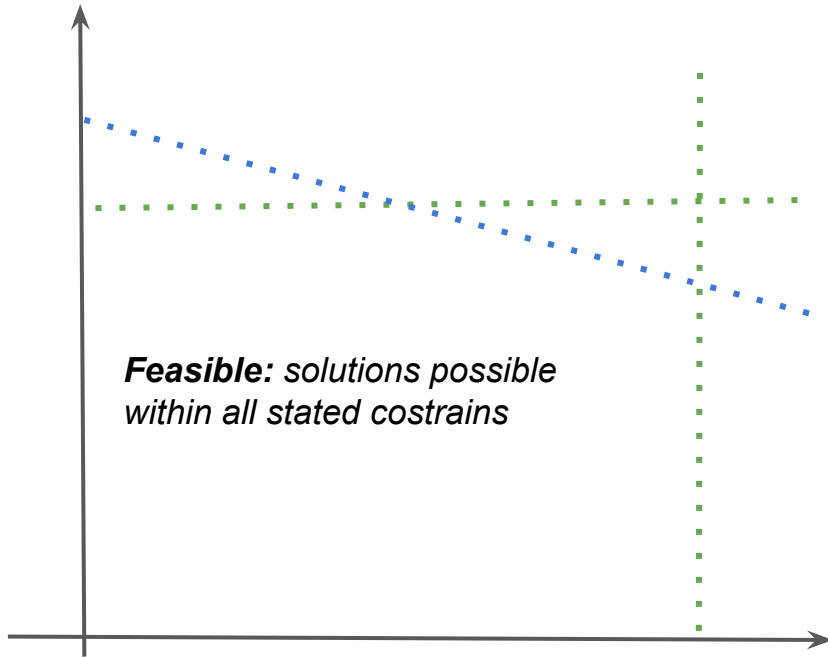
Additionally, the capacity for producing them simultaneously is limited by:

$$A + 2B < 120$$

The objective function is:

$$Z = 20A + 30B$$





The linear programming problem

Linear programming:

optimizing (maximize or minimize) a linear function subject to various constraints

Objective function

maximize $\mathbf{c}^T \mathbf{v}$

subject to

$$S\mathbf{v} = \mathbf{0}$$

Constraints

$$\mathbf{v}_i^{\text{lower}} \leq \mathbf{v}_i \leq \mathbf{v}_i^{\text{upper}} \quad \forall i \in \{1, \dots, r\}$$

Objective function

The **objective function** represents the biological goal or function that the cell is assumed to optimize.

It is a **linear function** of the reaction fluxes

Maximize (or minimize) $Z = \mathbf{c}^T \mathbf{v}$

Where:

- Z = objective value (scalar)
- $\mathbf{v}$ = vector of fluxes (rates of each reaction)
- $\mathbf{c}$ = vector of weights (same length as $\mathbf{v}$)
- Each c_i corresponds to how much reaction v_i contributes to the objective

Objective function

Minimize ATP production: This objective is stated to determine conditions of optimal metabolic energy efficiency.

Minimize nutrient uptake: This objective function is used to determine the conditions under which the cell will perform its metabolic functions while consuming the minimum amount of available nutrients.

Minimize redox production: This objective function finds conditions where the cells operate to generate the minimum amount of redox potential.

Minimize the Euclidean norm: This objective has been applied to satisfy the strategy of a cell to minimize the sum of the flux values, or to channel the metabolites as efficiently as possible through the metabolic pathways.

Maximize metabolite production: This objective function has been used to determine the biochemical production capabilities of *Escherichia coli*. In this analysis, the objective function was defined to maximize the production of a chosen metabolite (i.e. lysine or phenylalanine).

Maximize biomass and metabolite production: By weighing these two conflicting objectives appropriately, one can explore the tradeoff between cell growth and forced metabolite production in a producing strain.

Objective function

The Biomass Reaction:

To predict growth rate, we need to estimate the rate at which metabolites are converted to biomass constituents (e.g., nucleic acids, proteins, lipids)

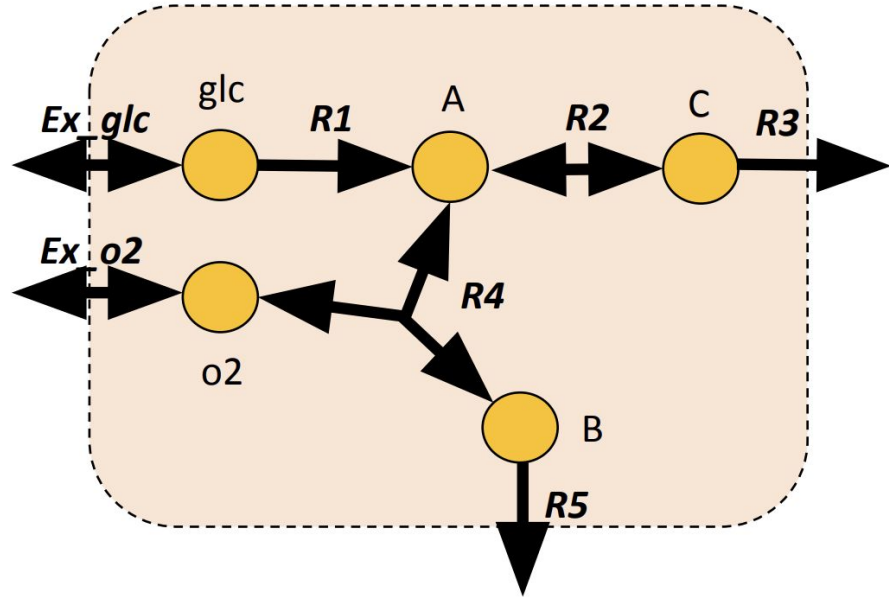
The “biomass reaction” predicts the exponential growth rate of the organism. Coefficients on metabolites are experimentally determined

Biomass reaction for *E. coli* iJO1366:

0.000223 10fthf[c] + 0.000223 2dmmql8[c] + 2.5e-005 2fe2s[c] + 0.000248 4fe4s[c] + 0.000223 5mthf[c] + 0.000279 accoa[c] + 0.000223 adocbl[c] + 0.49915 ala-L[c] + 0.000223 amet[c] + 0.28742 arg-L[c] + 0.23423 asn-L[c] + 0.23423 asp-L[c] + 54.12 atp[c] + 0.000116 bmocogdp[c] + 2e-006 btn[c] + 0.004952 ca2[c] + 0.000223 chor[c] + 0.004952 cl[c] + 0.000168 coa[c] + 2.4e-005 cobalt2[c] + 0.1298 ctp[c] + 0.000674 cu2[c] + 0.088988 cys-L[c] + 0.024805 datp[c] + 0.025612 dctp[c] + 0.025612 dgtp[c] + 0.024805 dttp[c] + 0.000223 enter[c] + 0.000223 fad[c] + 0.006388 fe2[c] + 0.007428 fe3[c] + 0.25571 gln-L[c] + 0.25571 glu-L[c] + 0.5953 gly[c] + 0.15419 glycogen[c] + 0.000223 gthrd[c] + 0.20912 gtp[c] + 48.7529 h2o[c] + 0.000223 hemeO[c] + 0.092056 his-L[c] + 0.28231 ile-L[c] + 0.18569 k[c] + 0.43778 leu-L[c] + 3e-006 lipopb[c] + 0.33345 lys-L[c] + 3.1e-005 malcoa[c] + 0.14934 met-L[c] + 0.008253 mg2[c] + 0.000223 mlthf[c] + 0.000658 mn2[c] + 7e-006 mobd[c] + 7e-006 mococdp[c] + 7e-006 mocogdp[c] + 0.000223 mql8[c] + 0.001787 nad[c] + 4.5e-005 nadh[c] + 0.000112 nadp[c] + 0.000335 nadph[c] + 0.012379 nh4[c] + 0.000307 ni2[c] + 0.012366 pe160[c] + 0.009618 pe161[c] + 0.004957 pe181[c] + 0.005707 pg160[c] + 0.004439 pg161[c] + 0.002288 pg181[c] + 0.18002 phe-L[c] + 0.000223 pheme[c] + 0.2148 pro-L[c] + 0.03327 ptrc[c] + 0.000223 pydx5p[c] + 0.000223 q8h2[c] + 0.000223 ribflv[c] + 0.20968 ser-L[c] + 0.000223 sheme[c] + 0.004126 so4[c] + 0.006744 spmd[c] + 9.8e-005 succoa[c] + 0.000223 thf[c] + 0.000223 thmpp[c] + 0.24651 thr-L[c] + 0.055234 trp-L[c] + 0.13399 tyr-L[c] + 5.5e-005 udcdp[c] + 0.1401 utp[c] + 0.41118 val-L[c] + 0.000324 zn2[c] + 0.008151 colipa[e] + 0.002944 clpn160[p] + .00229 clpn161[p] + 0.00118 clpn181[p] + 0.001345 murein3p3p[p] + 0.000605 murein3px4p[p] + 0.005381 murein4p4p[p] + 0.005448 murein4px4p[p] + 0.000673 murein4px4px4p[p] + 0.031798 pe160[p] + 0.024732 pe161[p] + 0.012747 pe181[p] + 0.004892 pg160[p] + 0.003805 pg161[p] + 0.001961 pg181[p]
-> 53.95 adp[c] + 53.95 h[c] + 53.9459 pi[c] + 0.74983 ppi[c]

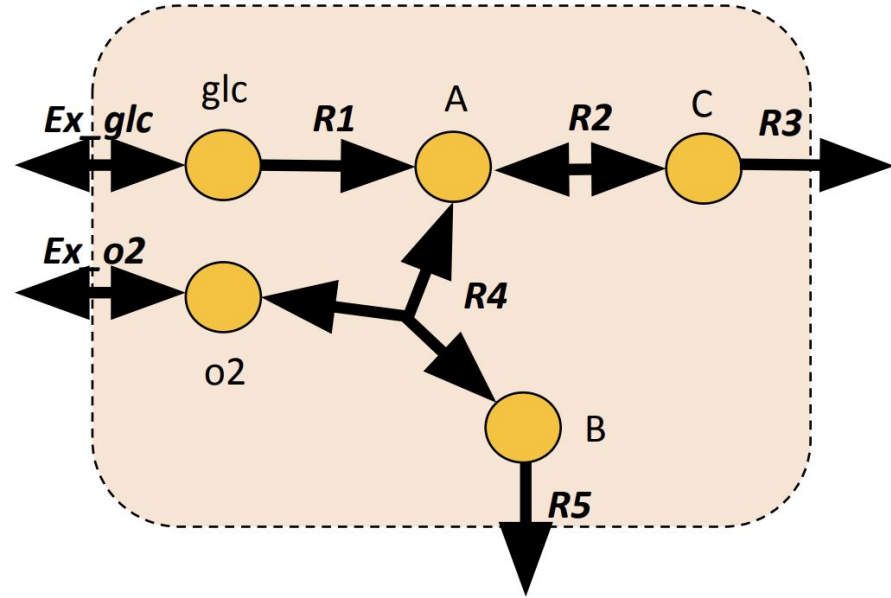
Example

Define c for the following system, where you want to maximize production of species C.



Example

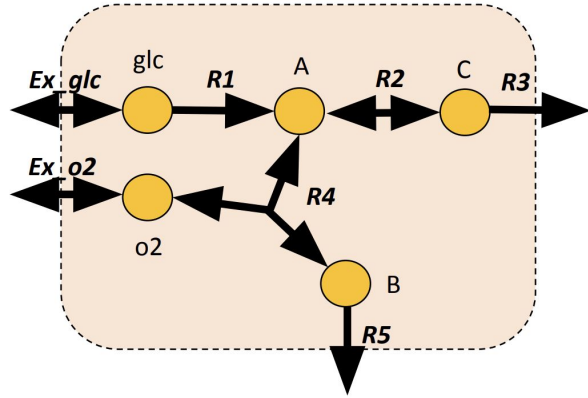
Define **c** for the following system, where you want to maximize production of species C.



Metabolite	Ex_glc	Ex_o2	R1	R2	R3	R4	R5
glc	-1	0	-1	0	0	0	0
o2	0	-1	0	0	0	-1	0
A	0	0	+1	-1	0	-1	0
C	0	0	0	+1	-1	0	0
B	0	0	0	0	0	+1	-1

Example

Define $\mathbf{c}$ for the following system, where you want to maximize production of species C.



maximize $\mathbf{c}^T \mathbf{v}$

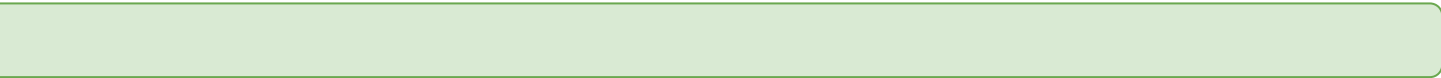
subject to

$$S\mathbf{v} = \mathbf{0}$$

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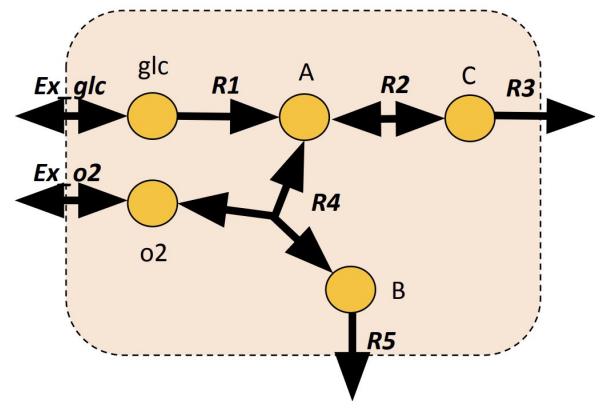
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glc	-1	0	-1	0	0	0	0
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A	0	0	+1	-1	0	-1	0
C	0	0	0	+1	-1	0	0
B	0	0	0	0	0	+1	-1

$$\mathbf{c} = [0, 0, 0, 1, 0, 0, 0]$$



Adjust S, c, lb, and ub to simulate:
a. anaerobic conditions ; b. aerobic conditions

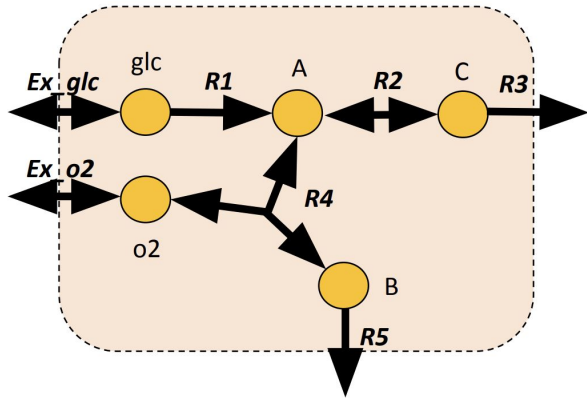
maximizing production of species C



Adjust S, c, lb, and ub to simulate:

a. anaerobic conditions ; b. aerobic conditions

maximizing production of species C



Anaerobic Conditions (No oxygen)

- Ex_o2 : no oxygen $\rightarrow lb = 0, ub = 0$

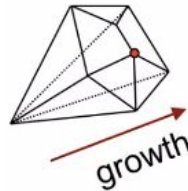
Aerobic Conditions (Oxygen available)

- Ex_o2 : oxygen allowed $\rightarrow lb = -20, ub = 0$

Linear programming

The **Simplex algorithm** is a method used to **solve linear programming (LP) problems**.

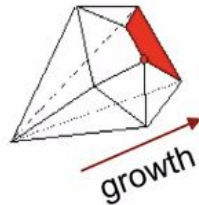
1. **Starts at a corner (vertex)** of the allowable solution space.
 2. **Walks along the edges** from vertex to vertex.
 3. At each step, it chooses the direction that **most increases the objective** (e.g., growth).
 4. **Stops when no step** can make the objective better.
- Guaranteed to find an optimal solution;
 - Exponential running time in worst case;
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Linear programming may result in multiple alternative solutions;

Alternative solutions represent different possible metabolic behaviors (through alternative pathways);

The solution space can be explored by various sampling and optimization methods.

Parsimonious Flux Balance Analysis

Parsimonious Flux Balance Analysis (**pFBA**) is an **extension of standard FBA** that not only finds a solution that **maximizes the objective function** (e.g., growth), but also **minimizes the total flux** through the network.

Two steps:

1. Standard FBA

- Maximize the objective (e.g., biomass production)
- Find the optimal value Z^*

2. Minimize total flux subject to achieving Z^*

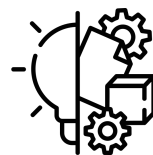
$$\text{minimize} \quad \sum_i |\mathbf{v}_i|$$

subject to

$$S\mathbf{v} = \mathbf{0}$$

$$\mathbf{v}_i^{\text{lower}} \leq \mathbf{v}_i \leq \mathbf{v}_i^{\text{upper}} \quad \forall i \in \{1, \dots, r\}$$

$$\mathbf{c}^T \mathbf{v} \geq \gamma Z^* \quad \gamma \in [0, 1]$$



Because **cells tend to be efficient** — they prefer to use the **fewest and simplest** pathways to achieve their goals (like growth), minimizing enzyme usage, energy, and resource investment.

Flux Variability Analysis

Flux Variability Analysis (**FVA**) determines the **range of possible fluxes** (minimum and maximum) for each reaction in a metabolic network **while still allowing the model to achieve a given objective**

FBA:

- We find **one** flux distribution that optimizes the objective.
- But **many alternative solutions** can achieve the same objective value.

FVA reveals this flexibility:

- It tells you **how tightly or loosely constrained** each reaction is.
- It helps distinguish between **essential** and **alternative** pathways.

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- It helps distinguish between **essential** and **alternative** pathways.

After performing FBA to get the optimal objective value Z^* , for each reaction v_i , FVA solves **two LP problems**:

1. **Minimize** v_i subject to:

$$S\mathbf{v} = 0$$
$$\mathbf{v}_i^{\text{lower}} \leq \mathbf{v}_i \leq \mathbf{v}_i^{\text{upper}}$$
$$\mathbf{c}^T \mathbf{v} \geq \gamma Z^* \quad \gamma \in [0, 1]$$

2. **Maximize** v_i , with the same constraints.

Repeat this for **every reaction** to get a **range**: $v_i^{\text{min}} \leq v_i \leq v_i^{\text{max}}$

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FVA reveals this flexibility:

- It tells you **how tightly or loosely constrained** each reaction is.
- It helps distinguish between **essential** and **alternative** pathways.

FVA shows **all possible solutions** that give the same objective.

Helps interpret **metabolic flexibility, redundancy, and robustness**.

Duality

In linear programming, **every optimization problem** (the "Primal") **has an associated "Dual" problem**.

1. The **Primal** problem is the one you're solving (e.g., maximize biomass).

$$\begin{aligned} &\text{maximize} && \mathbf{c}^T \mathbf{v} \\ &\text{subject to} && \\ &&& S\mathbf{v} = \mathbf{0} \\ &&& \mathbf{v}_i^{\text{lower}} \leq \mathbf{v}_i \leq \mathbf{v}_i^{\text{upper}} \quad \forall i \in \{1, \dots, r\} \end{aligned}$$

2. The **Dual** problem provides **economic or biological interpretation** of the constraints.

It corresponds to finding **shadow prices** for the constraints, reflecting how **tight or valuable** each constraint is.

Each metabolite has its "**shadow price**" λ_i

$\lambda \in \mathcal{R}^m$: dual variables for mass balance constraints

$\gamma^+, \gamma^- \in \mathcal{R}^n$: duals for lower and upper bounds

$$\begin{aligned} \min_{\lambda, \gamma^-, \gamma^+} & ub^T \gamma^+ - lb^T \gamma^- \\ \text{subject to:} & S^T \lambda + \gamma^+ - \gamma^- \geq c \\ & \gamma^+, \gamma^- \geq 0 \end{aligned}$$

Omics Integration

Integrating **omics data** into **FBA** allows metabolic models to become more **context-specific**, **dynamic**, and **biologically realistic**.

Omics Type	Description	Role in FBA
Transcriptomics	mRNA abundance	Indirect proxy for enzyme presence
Proteomics	Protein abundance	More accurate for enzyme activity
Metabolomics	Metabolite concentrations	Help constrain fluxes and directions
Fluxomics	Measured reaction rates	Validate or constrain model fluxes

Transcriptomics

Each **reaction** in a GEM is linked to a **gene set** through Boolean logic:

- **AND**: multiple genes are needed (e.g., multi-subunit enzymes)
- **OR**: any of several genes can support the reaction (isoenzymes)

$$GSE_i = \min(expression_{geneA}, expression_{geneB})$$

$$GSE_i = expression_{geneA} + expression_{geneB}$$

These GPR rules translate into a **Gene Set Expression (GSE)** level for each reaction, which drives how much **flux is allowed** through the reaction.

$$h(GSE_i) = \begin{cases} (1 + |\log(GSE_i)|)^{\frac{GSE_i-1}{|GSE_i-1|}} & \text{if } GSE_i \in \mathbb{R}^+ \setminus \{1\} \\ 1 & \text{if } GSE_i = 1 \end{cases} \quad \longrightarrow \quad v_i^{\text{Low}} h(GSE_i) \leq v_i \leq v_i^{\text{Up}} h(GSE_i)$$

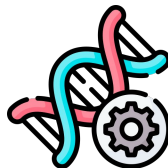
If **GSE $\approx$ 0** $\rightarrow$ $h(\text{GSE})$ becomes very negative $\rightarrow$ **lowers allowable flux**

If **GSE $\approx$ 1** $\rightarrow$ $h(\text{GSE})=1$ $\rightarrow$ **default bounds retained**

If **GSE $\gg$ 1** $\rightarrow$ scales up

Other approaches

GIMME (Becker & Palsson, 2008)



- Keeps reactions needed for an objective (like growth)
- Removes reactions with low expression
- Optimizes to be consistent with both growth and expression



Diet Constraints

Diet and media constraints define the availability of nutrients that an organism can take up from its environment. In metabolic models, this is done by setting bounds on exchange reactions, which control how metabolites enter or exit the system.

Nutrient	Exchange Reaction	Unit
Glucose	EX_glc__D_e	mmol/gDW/h
Oxygen	EX_o2_e	mmol/gDW/h
Amino acids	EX_ile__L_e, EX_arg__L_e, etc.	mmol/gDW/h
Fatty acids	EX_hdca_e, etc.	mmol/gDW/h
Vitamins	EX_thm_e, EX_cb11_e	mmol/gDW/h

Exchange Reactions

- Define uptake/secretion from the environment
- Apply **upper/lower bounds** to simulate a **diet or growth medium**



models



fluxes



algorithms



509

contributors

56

release

v0.29.1

cobrapy is a python package that provides a simple interface to metabolic constraint-based reconstruction and analysis.

COBRApy - Constraint-Based Reconstruction and Analysis in Python



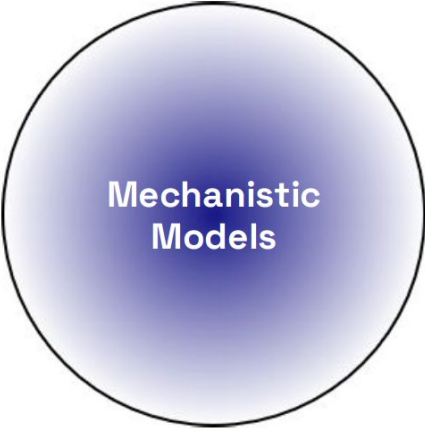
What is COBRApy?

COBRA methods are widely used for genome-scale modeling of metabolic networks in both prokaryotes and eukaryotes. **COBRApy** is a constraint-based modeling package that is designed to accommodate the biological complexity of the next generation of COBRA models and provides access to commonly used COBRA methods, such as flux balance analysis, flux variability analysis, and gene deletion analyses.

Our aim with COBRApy is to provide useful, efficient infrastructure for:

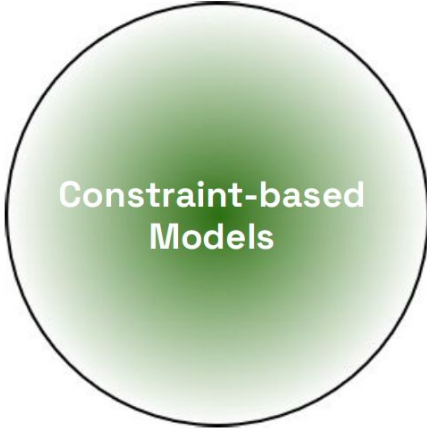
- creating and managing metabolic models
- accessing popular solvers
- analyzing models with methods such as FVA, FBA, pFBA, MOMA etc.
- inspecting models and drawing conclusions on gene essentiality, testing consequences of knock-outs etc.

Hybrid Modeling



Mechanistic Models

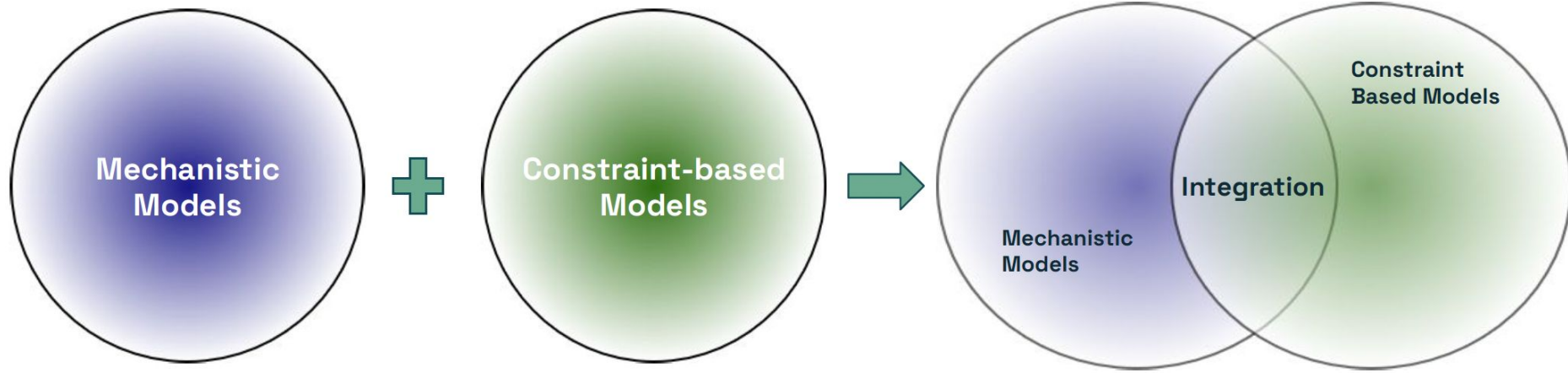
- Detailed mathematical equations
- Captures temporal dynamics
- Step-by-step system interactions
- Computationally expensive simulations



Constraint-based Models

- Applies biological constraints
- Restricts possible states
- Emphasizes efficiency and optimization
- Explores feasible system outcomes

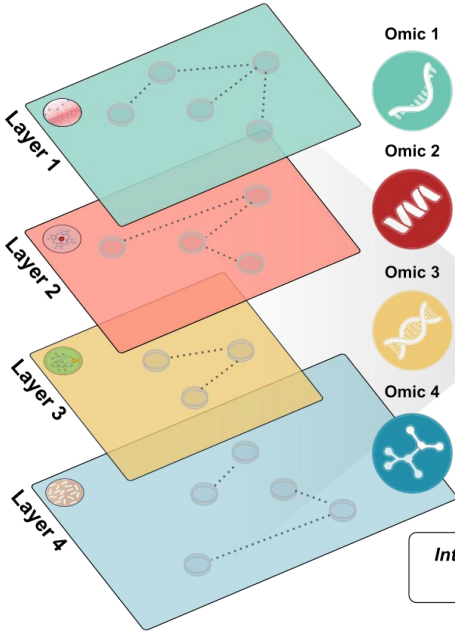
Hybrid Modeling



- **Combines dynamic details with system-wide constraints**
- **Provides a comprehensive view of biological systems**
- **Balances precision with efficiency**

UnifiedGreatMod

Data integration
Connecting data from multiple
Omics measurements



Omic 1



Omic 2



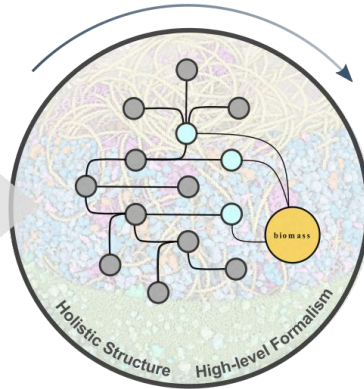
Omic 3



Omic 4



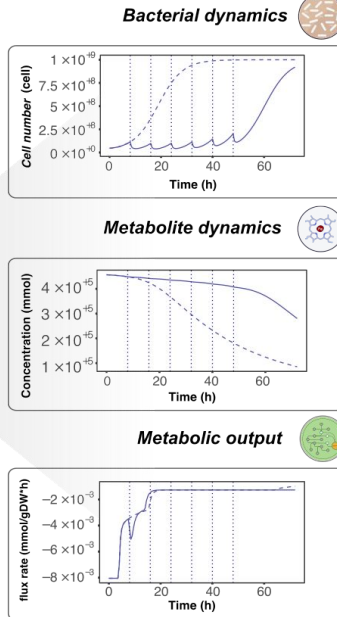
**Unified Model
harmonizing layers**



*Interacting domains of
knowledge*

*Computational translation
and modelling*

Model Simulation
Omics-driven predictive modelling



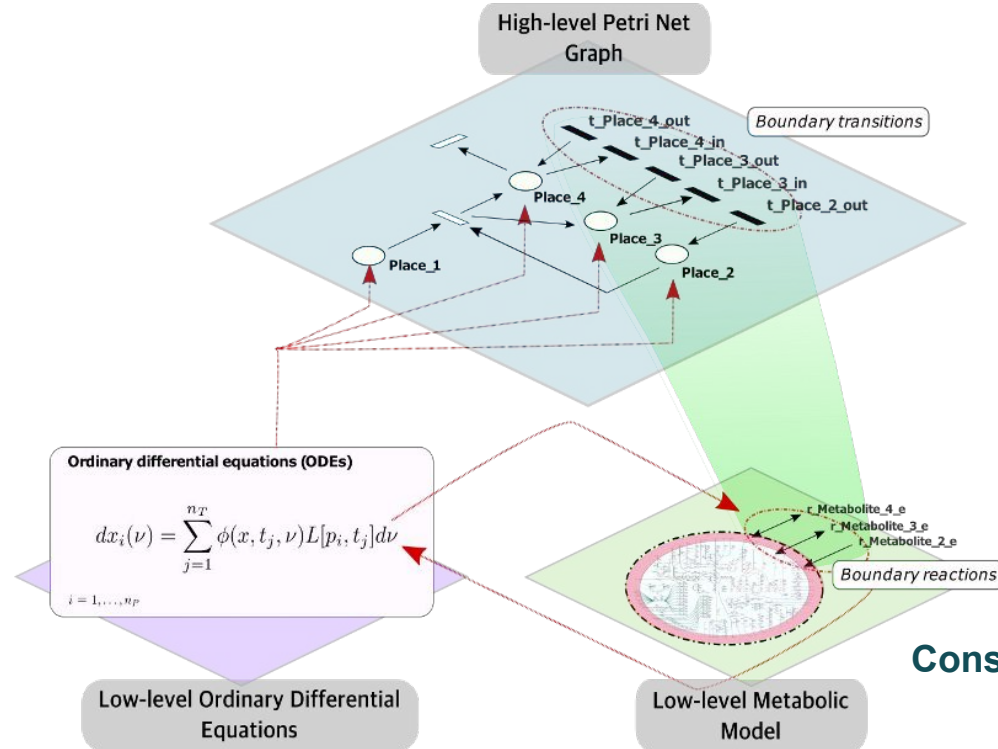
*Harmonization is the central
aspect of
UnifiedGreatMod.*

The harmonization, a central aspect of our paradigm, facilitates a dialogue among biological data and merges the computational aspects of the paradigm.

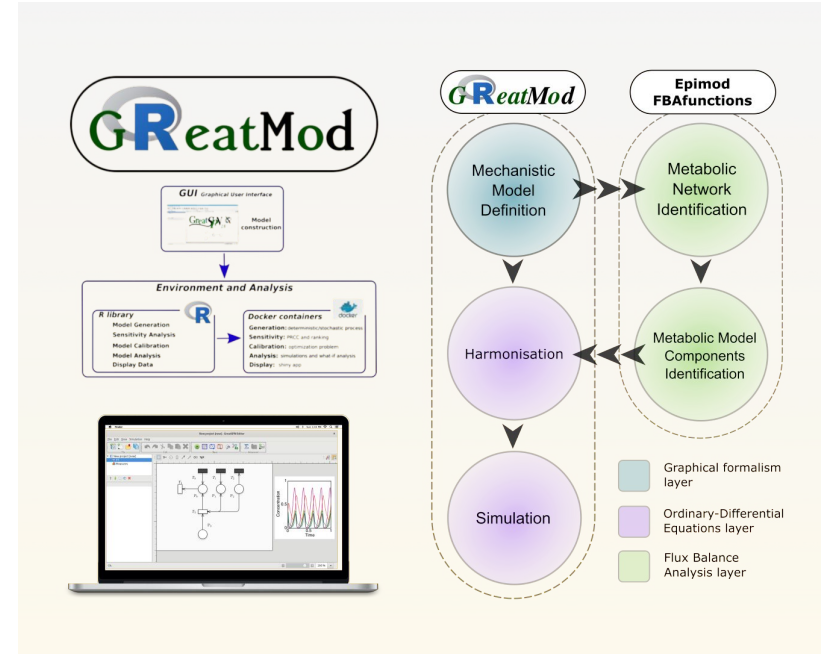
The combination of mechanism and constraint-based models simulate the entire system to yield temporal evolutions of key biological entities.

Modeling paradigm

Graphical Model (Petri Net)

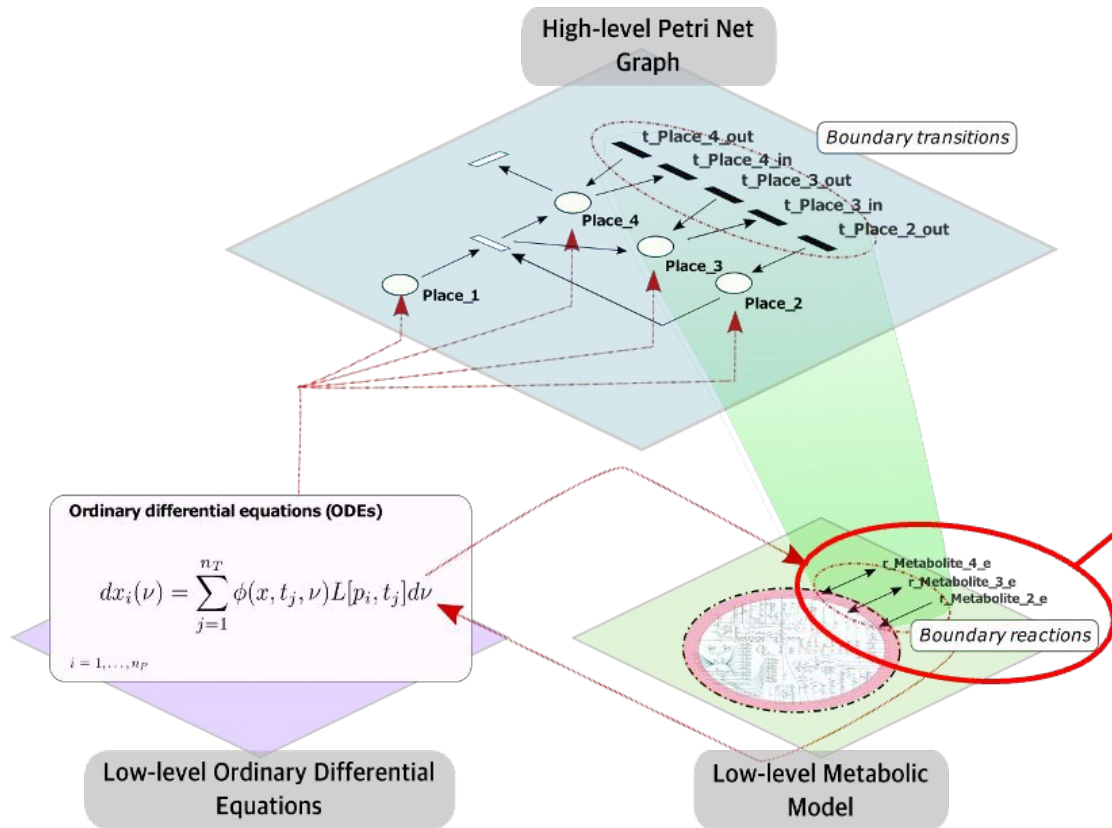


Dynamic/Mechanistic Model (ODEs)



Constraint-Based Model (FBA)

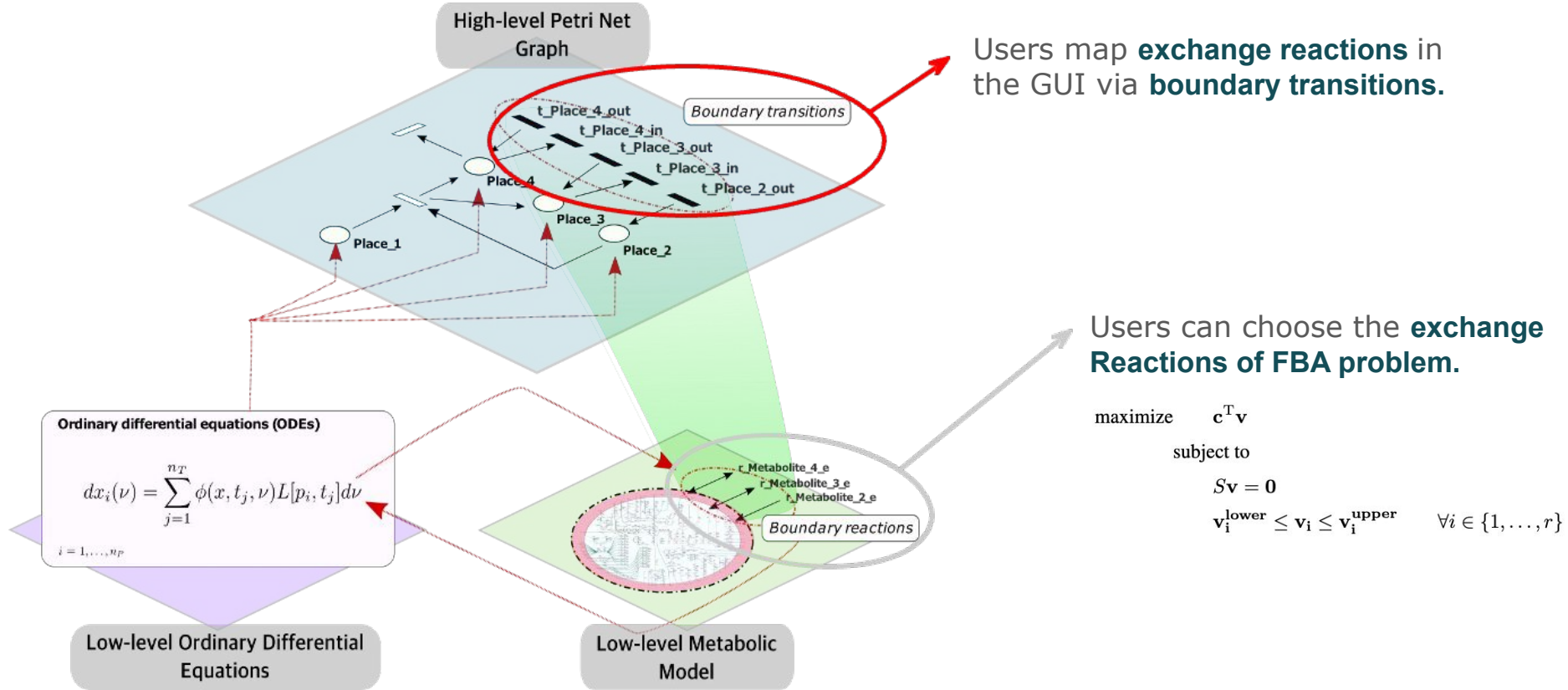
Modeling paradigm



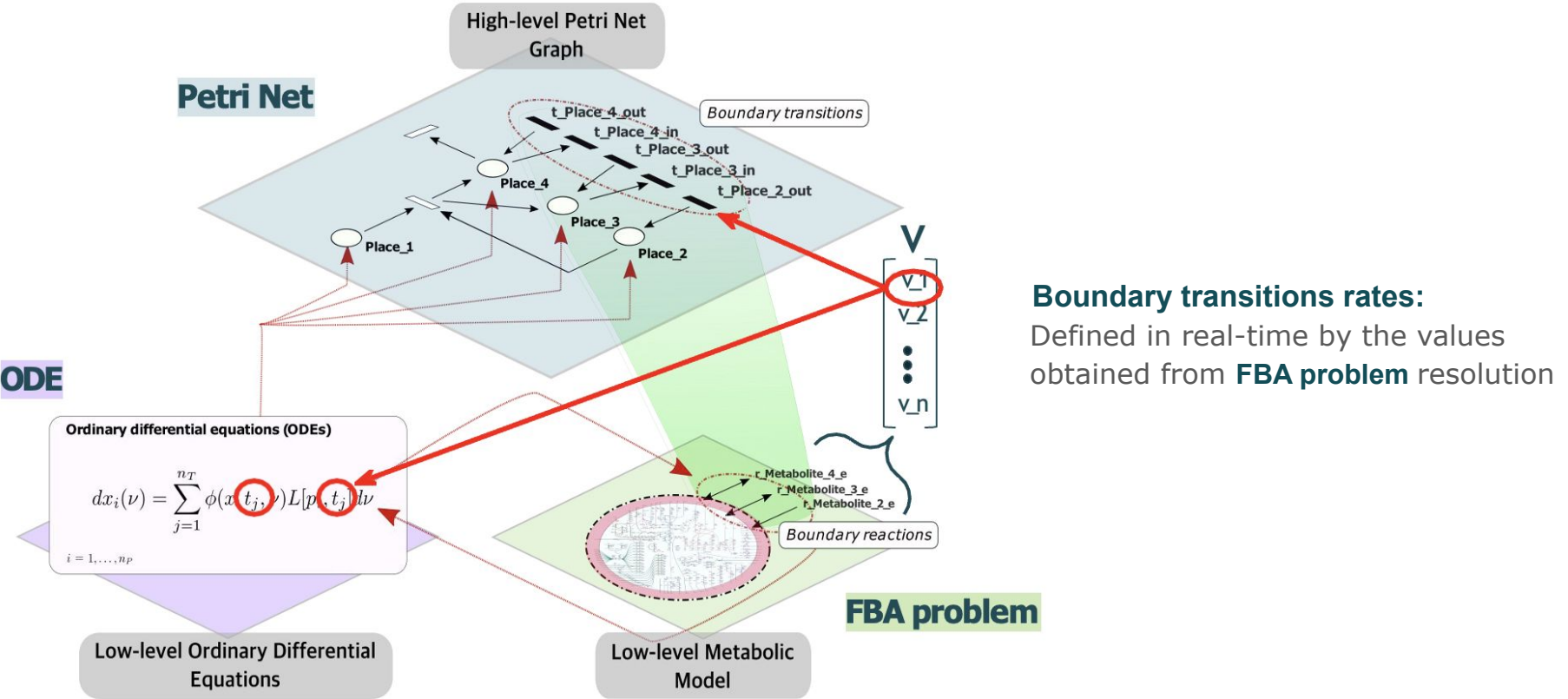
Users can choose the **exchange Reactions of FBA problem.**

$$\begin{aligned}
 &\text{maximize} && \mathbf{c}^T \mathbf{v} \\
 &\text{subject to} && \\
 &&& S\mathbf{v} = \mathbf{0} \\
 &&& \mathbf{v}_i^{\text{lower}} \leq \mathbf{v}_i \leq \mathbf{v}_i^{\text{upper}} \quad \forall i \in \{1, \dots, r\}
 \end{aligned}$$

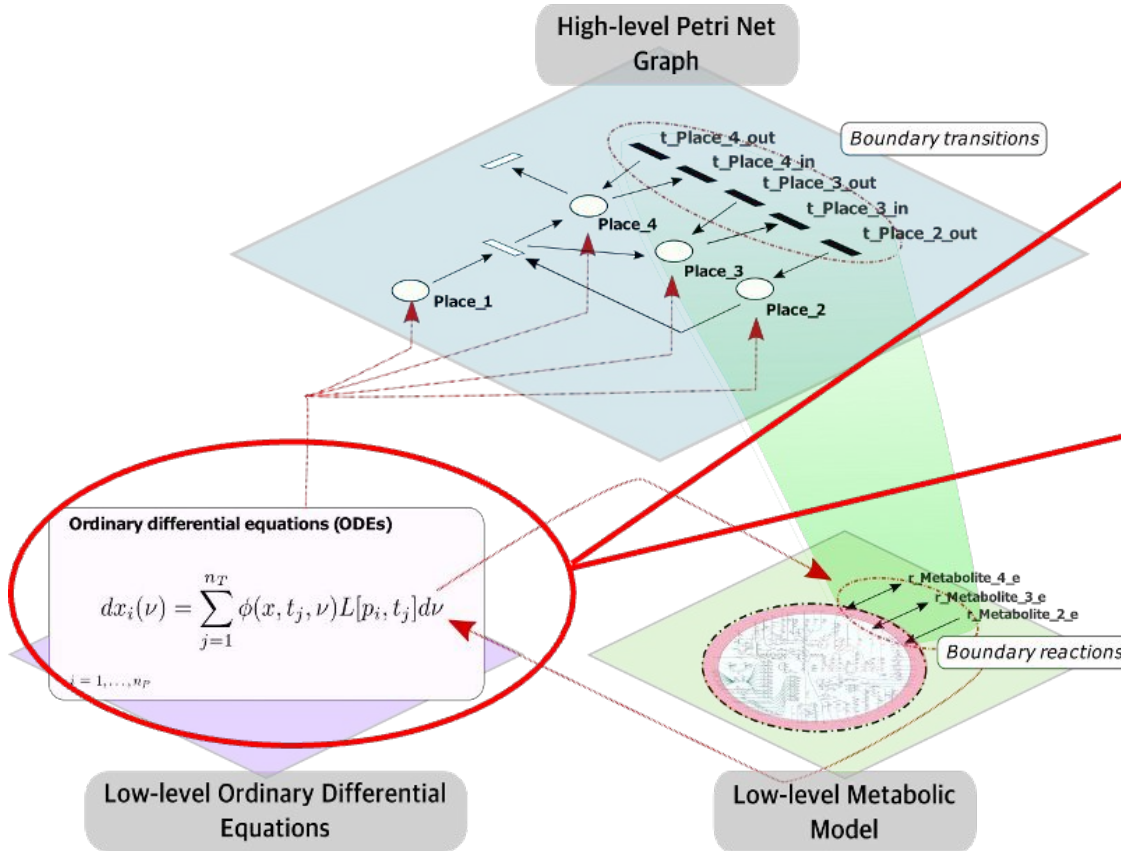
Modeling paradigm



Modeling paradigm

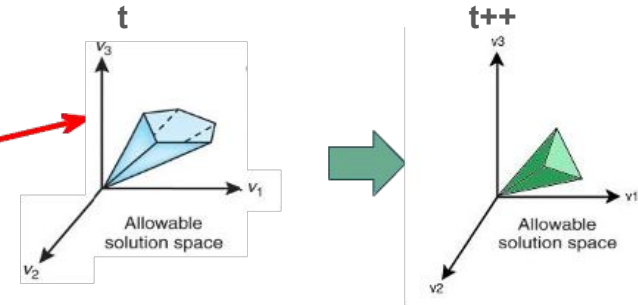


Modeling paradigm



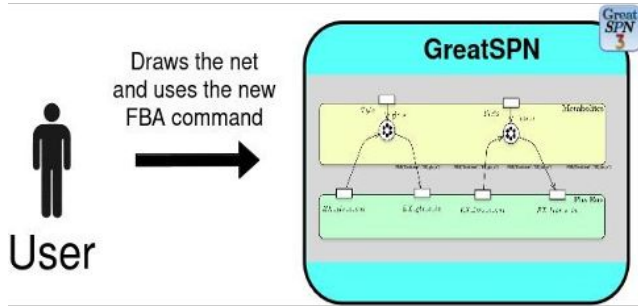
FBA constraints:

Determined dynamically by the system's state (ODEs) at a given time t .



$$\begin{aligned}
 &\text{maximize} && \mathbf{c}^T \mathbf{v} \\
 &\text{subject to} && \\
 &&& \mathbf{S} \mathbf{v} = \mathbf{0} \\
 &&& \mathbf{v}_i^{\text{lower}} \leq \mathbf{v}_i \leq \mathbf{v}_i^{\text{upper}} \quad \forall i \in \{1, \dots, r\}
 \end{aligned}$$

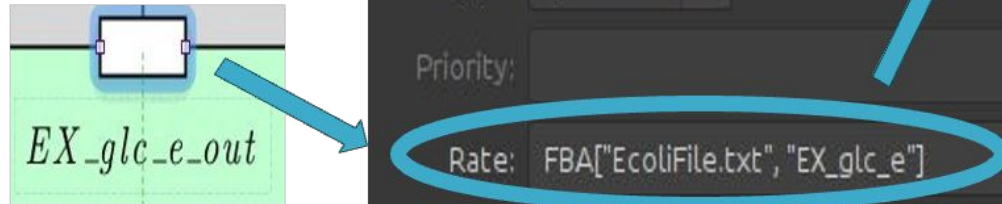
Modeling paradigm



Command Parameters:

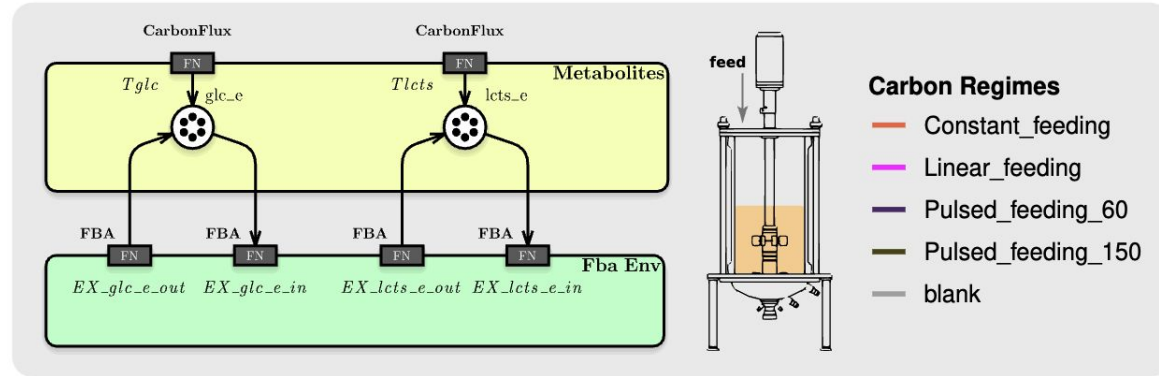


1. **FBA problem File Name**,
2. **Reaction ID**,
3. **Multiplicative Constant** (default: 1),
4. **Name of the place corresponding to this number of bacteria**
5. **Name of the biomass place**
6. **Flag for the biomass transition** (default is FALSE)



Case study

E. coli batch growing model according to the supplementation scenario over time.



An optimal test bed for UnifiedGreatMod is given by the simulation of the **carbon catabolite repression phenomenon** exhibited by *E. coli*. A regulatory mechanism by which the expression of genes required for the utilization of secondary sources of carbon is prevented by the presence of a preferred substrate. **The bacterium adapts its metabolism to the environment's available carbon sources** (i.e., D-glucose or lactose).

The LacZ system plays a crucial role in *E. coli*'s adaptive response to changes in available carbon sources. The role is to ensure that the cell expends energy on producing enzymes encoded by the lac operon only when necessary.

Case study

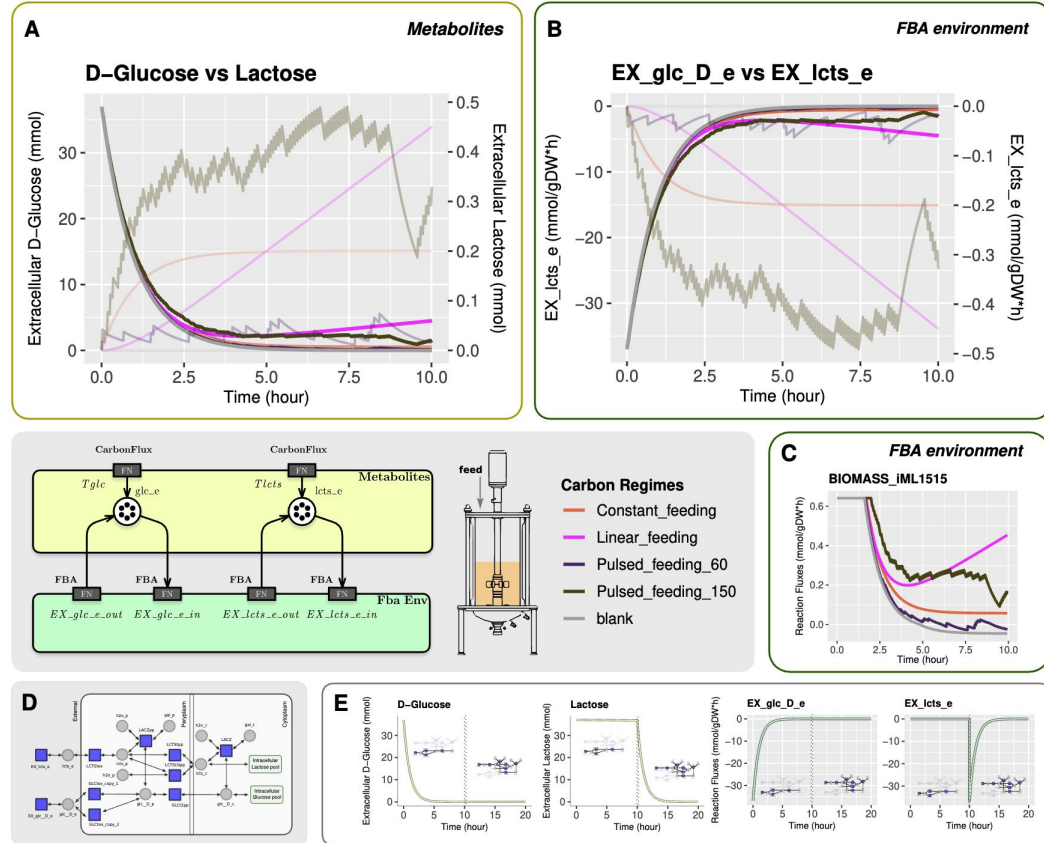
(A) Profiles representing the time-course changes in the concentrations of D-glucose and lactose under different conditions of carbon administration.

(B) FBA-estimated metabolic fluxes for D-glucose and lactose exchanges at time resolution.

(C) FBA-estimated E.coli biomass objective function at time resolution.

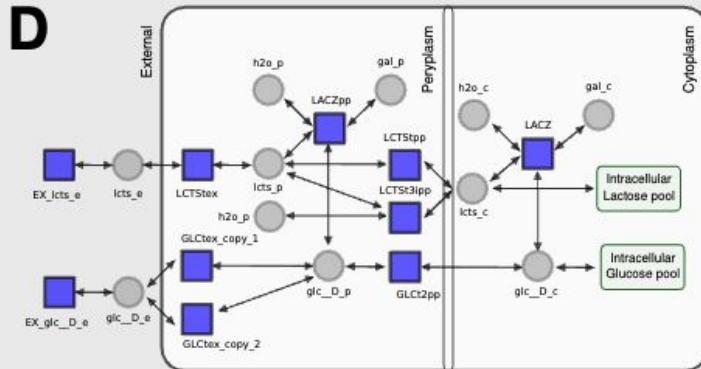
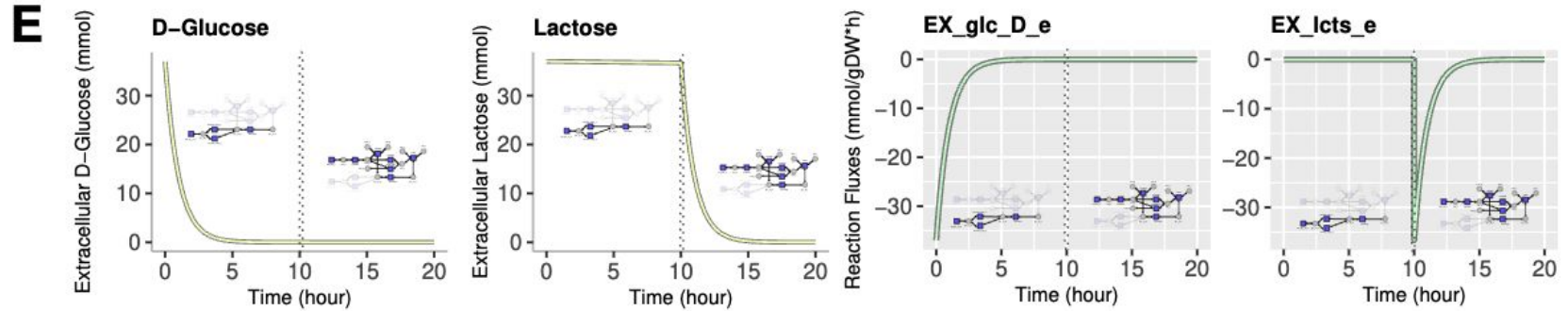
(D) Graph of the subnetwork from iML1515 for the uptake of D-glucose and Lactose and components encoded by the lac operon. Grey circles represent the metabolites from the subnetwork, and blue squares represent the subsystem reactions. The pathway starts with the transport of D-Glucose and Lactose from the extracellular space to the periplasm, where lactose can be transformed into periplasmatic D-Glucose (glc_D_p) through a reaction from periplasmatic beta-galactosidase. D-glucose and lactose can also be directed to the cytoplasm where beta-galactosidase cleaves lactose into D-glucose.

(E) Results collectively highlight the role of the LacZ system in enabling the switch from D-glucose to lactose metabolism when necessary.



Case study

Results collectively highlight the role of the LacZ system in enabling the switch from D-glucose to lactose metabolism when necessary.



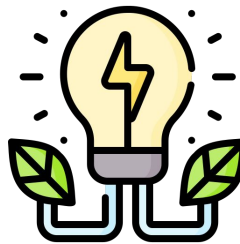
Graph of the subnetwork from iML1515 for the uptake of D-glucose and Lactose and components encoded by the lac operon. Grey circles represent the metabolites from the subnetwork, and blue squares represent the subsystem reactions. The pathway starts with the transport of D-Glucose and Lactose from the extracellular space to the periplasm, where lactose can be transformed into periplasmatic D-Glucose (glc_D_p) through a reaction from periplasmatic beta-galactosidase. D-glucose and lactose can also be directed to the cytoplasm where beta-galactosidase cleaves lactose into D-glucose.

Conclusion: Flux Balance Analysis in a Nutshell

FBA is a powerful constraint-based modeling approach that allows the prediction of metabolic fluxes at steady state **without requiring kinetic parameters**.

Enables analysis of:

- **Growth prediction** via **biomass objective**
- **Flux distribution** under different conditions
- **Gene knockouts** and metabolic capabilities
- **Integration of omics data** (transcriptomics, proteomics, diet)



Take away:

FBA abstracts the complex biochemistry of the cell into a solvable mathematical problem, giving biological insight into metabolism, growth, and gene-function relationships — especially at the genome scale.

Conclusions

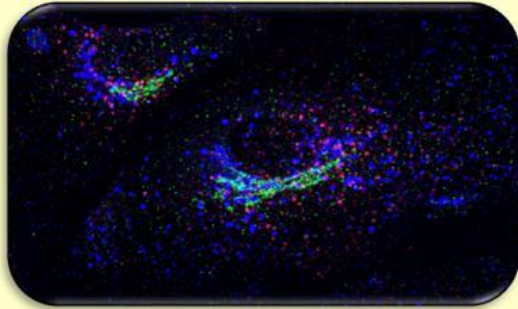
Cellular Biology approach

Immunofluorescence

Live imaging

Western Blot

Real-time PCR



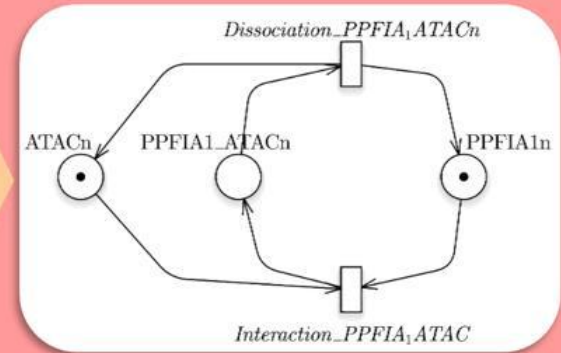
Understanding at fine molecular detail the components involved in the pathway

Data - Driven models

Computational Biology approach

Petri Nets formalism

Dynamic & Constraints models



Viewing the system as **a whole** and investigating its characteristics